# Welfare Analysis of Changing Notches: Evidence from Bolsa Família*

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*We develop a framework to bound the welfare impacts of arbitrarily large reforms to notches using two sufficient statistics: the change in the number of households locating below the new notch and the change in the number of households locating below the old notch. Our bounds hold in a wide class of models, highlighting a new way to use reduced-form bunching evidence for welfare analysis without making strong assumptions on the economic environment. We estimate these two statistics using a difference-in-difference strategy for a reform to the anti-poverty program Bolsa Família, finding that the reform's MVPF is between 0.90-1.12.*

*JEL: H31, H53, I38, O12*
*Keywords: bunching, jumping, notches, MVPF, bounds, sufficient statistics*

Throughout both developed and developing countries, tax and transfer schedules often feature notches wherein incremental changes in behavior lead to discrete changes in benefits or tax liabilities (Slemrod (2010); Kleven (2016)). For example, Saving and Viard (2021) document numerous notches in the United States tax code including the well-known Medicaid eligibility notch (Yelowitz, 1995), Kleven and Waseem (2013) document notches of up to 4 percentage points in average personal income tax rates in Pakistan, and Bachas and Soto (2021) document 10 percentage point jumps in the average corporate tax rate in Costa Rica. Given this prevalence, it is important for economists and policymakers to understand the welfare impacts of policy reforms that change the structure of notches in tax and transfer systems.

Typically the behavioral and welfare impacts of notches are analyzed using the standard bunching approach whereby one estimates the mass of agents bunching around a notch

* Some of the ideas in this paper were part of a chapter of Juan's dissertation at Stanford University, titled "Welfare Analysis of Transfer Programs with Jumps in Reported Income: Evidence from the Brazilian Bolsa Família". We are grateful to members of Juan's dissertation committee Douglas Bernheim, Raj Chetty, Luigi Pistaferri and Florian Scheuer for their advice and guidance. We also want to thank Pierre Bachas, Jose Maria Barrero, Alexander Gelber, David McKenzie, Berk Özler, Peter Phillips, and seminar participants at the NBER Public Economics Meeting, Stanford University, Tulane University, the World Bank, the University of Auckland, Brookings, and IIPF for their comments and suggestions. Thanks to four anonymous reviewers as well as the editor, Kory Kroft, whose comments all helped improve the paper. This research was supported in part using high performance computing (HPC) resources and services provided by Technology Services at Tulane University, New Orleans, LA.

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and then translates this bunching mass into a structural parameter(s) (Kleven, 2016). One can then use these structural parameters to gauge the welfare impacts of previous or proposed policy changes, although in this literature they are more commonly used to identify alternative schedules that will improve welfare. However, the chief limitation of this approach is that translating a bunching mass into a structural parameter typically relies heavily on modeling assumptions. For example, assumptions on household preferences, the types of behavioral responses available to households, and the degree of optimization frictions can all greatly impact the mapping between a bunching mass and a structural parameter.[1] Consequently, Kleven (2016) states "although bunching provides compelling non-parametric evidence of a behavioral response, moving from observed bunching to a structural parameter that can be used to predict the effects of policy changes is difficult."

This paper develops a method that uses changes in bunching to form bounds for the welfare impacts of notch reforms. In particular, we show how to bound the welfare impact of notch reforms in transfer schedules using two sufficient statistics: (1) the "jumping effect", which captures the change in the number of households locating at or below the new notch as a result of the reform (e.g., households jumping down to bunch at the new notch) and (2) the "bunching effect", which captures the change in the number of households locating at or below the old notch as a result of the reform (e.g., initially bunching households that move from the old notch towards the new notch). Importantly, our bounds require only minimal structure on the household problem: households can have arbitrary preference heterogeneity, households can have any number of choice variables (and, therefore, can respond to the reform in various ways), and households may face optimization frictions such as limited choice sets and/or adjustment costs. Thus, our approach overcomes an important limitation of the standard bunching approach by using reduced-form bunching evidence to inform welfare analysis without making strong assumptions about household preferences (e.g., our framework does not require a quasi-linear isoelastic utility function), the types of behavioral responses available to households (e.g., our framework allows for labor supply responses as well as misreporting responses), or the type/magnitude of optimization frictions. However, our approach can only be used to analyze the impacts of observed reforms whereas the standard bunching approach recovers structural parameters that can be used to analyze hypothetical reforms or recover the optimal policy; thus, our approach should be viewed as *complementary* to the standard bunching approach.

We then apply our method to bound the welfare impact of a reform that expanded a notch in one of the world's largest cash transfer programs, Bolsa Família (BF). BF, a means-tested anti-poverty program in Brazil, is a particularly interesting program to study for two reasons: (1) BF is one of the few cash transfer programs for which eligibility is based on self-reported household income (Fruttero, Leichsenring and Paiva, 2020), and (2) the benefit schedule features a pronounced eligibility threshold - a notch - wherein households are eligible for benefits only if they report an income below this threshold. Given the large portion of Brazil's workforce in the informal sector, these two features generate substantial

[1] This is not always the case: in some applications, structural parameters appear robust to model augmentations, e.g., Best et al. (2020).

scope for income misreporting. We study a 2014 reform to BF in which the government increased both the eligibility threshold and the amount of money given to beneficiary households. Using longitudinal administrative data, we find strong evidence of changes in bunching in response to this reform. Plugging our empirical estimates into our sufficient statistics framework, we find that the reform was welfare improving if the government values giving R$0.90 to BF recipients more than spending R$1 on their next best alternative and welfare decreasing if the government values giving R$1.12 to BF recipients less than spending R$1 on their next best alternative. Because BF has high coverage of very poor households, we argue that it is highly likely that the government is willing to spend R$1 to get R$0.90 to BF recipients. Thus, despite strong evidence of behavioral responses to the reform, our findings indicate that these responses are unlikely to be large enough to generate efficiency costs that outweigh the equity benefits of increased program generosity. Our findings highlight the importance of having a theoretical framework that translates behavioral responses to social programs into welfare impacts, a point recently emphasized in Gerard and Gonzaga (2021).

The paper begins by building intuition in a simple, static model in which a government provides a constant transfer only to households who report an income below a certain threshold; hence, the transfer schedule features a notch. Households are endowed with an income and choose how much income to report to the government subject to misreporting costs. We derive bounds on the MVPF of changing the notch, comprising both an increase in the benefit level and eligibility threshold. The MVPF is the ratio of households' willingness-to-pay (WTP) for the reform relative to the government's budgetary cost of the reform (Hendren and Sprung-Keyser, 2020). We show that aggregate WTP can be bounded using two empirical objects: a lower bound can be formed with an estimate of the "jumping effect", which captures the change in the number of households locating at or below the new notch as a result of the reform, and an upper bound can be formed with an estimate of the "bunching effect", which captures the change in the number of households locating at or below the old notch as a result of the reform. In this simple model, the "jumping effect" consists of households who jump down to bunch precisely at the new notch and the "bunching effect" is comprised of households who initially bunch precisely at the old notch and move towards the new notch. For households who do not jump or bunch, their WTP is simply equal to the amount of additional money they receive. However, for jumping and bunching households, their WTP differs from their increase in benefits; this is because we cannot appeal to the envelope condition given we consider arbitrary discrete reforms. Instead, we use revealed preference arguments to bound the amount that these households are willing to pay. We can therefore bound aggregate WTP and, in turn, the MVPF provided we can estimate the jumping effect and the bunching effect. Because the welfare impact of a reform is equal to the MVPF multiplied by a normative welfare weight, our bounds on the MVPF enable us to make welfare statements about the impacts of changing a notch using estimable, reduced-form objects.

While our results are initially stated in the context of a simple, static misreporting model to build intuition, we show that our welfare bounds are highly robust to the household problem. In particular, our bounds hold in models with any sort of behavioral response margin (e.g., households can respond to the reform via labor supply responses instead of misreporting responses), arbitrary preference heterogeneity, adjustment costs, limited choice sets (e.g., households face labor supply frictions), and misperceptions of the benefit schedule (as long as household behavioral responses to the reform increase utility on average). The key intuition for this generality comes from revealed preference: we derive welfare bounds by calculating the impacts of notch reforms assuming no one re-optimizes (by revealed preference, behavioral responses can only improve welfare). Hence, our bounds are robust to the types of behavioral responses available to households along with restrictions on household choice sets precisely because behavioral responses are held fixed when formulating our bounds. In the context of this more general framework, the "jumping effect" and "bunching effect" are still the sufficient statistics however the "jumping effect" comprises not only households that jump down to bunch precisely at the new threshold but also households that jump down anywhere below the new notch and the "bunching effect" comprises not only the reduction in bunching at the old notch but also any other change in the number of households anywhere below the old notch. Intuitively, in a world with frictions the "jumping effect" and "bunching effect" must also account for changes strictly below the new and old notches, respectively, due to diffuse bunching behavior. Moreover, we discuss how our bounds can be augmented to allow for dynamic decision making with uncertainty and for more complex policy environments. We also show that our bounds are sharp: they cannot be tightened without making additional structural assumptions. We therefore refer to the jumping effect and bunching effect as sufficient statistics to bound the welfare impact of a notch change. We view this generality as a key contribution of our framework as not only does it allow for our framework to be adapted to analyze notch reforms in other transfer programs, but it also highlights a new way to use reduced-form bunching evidence to inform welfare analysis without making strong assumptions on the household problem.

We then turn our attention to the welfare impact of the June 2014 Bolsa Família reform. Our main empirical analysis restricts to single-adult households with no children.[2] Prior to June 2014, single-adult households with no children were eligible for a monthly unconditional benefit of R\$70 if and only if they reported an income below the extreme-poverty threshold of R\$70 per-capita per-month.[3] In June 2014, both the eligibility threshold and benefit were raised by 10%. We have access to administrative data spanning December 2011 to September 2016 from Cadastro Único, which is the Brazilian government's national registry used to determine eligibility for all federal social welfare programs. Using this data, we seek to estimate the bunching effect along with the jumping effect. The key identification challenge is therefore to measure how the number of households reporting

[2]Single-adult households with no children make up 6.3% of BF beneficiaries in June 2014. We also repeat our empirical analysis for two-adult households with no children (who make up 21.1% of BF beneficiaries) and find very similar welfare bounds.

[3]BF also includes a conditional variable benefit for households with children. To receive this conditional benefit, households must make specified health and education investments in their children and have a reported per-capita income below the "poverty threshold" (which is higher than the extreme-poverty threshold).

incomes at and below the old and new notch would have evolved in the post-reform period had the reform not occurred. Our identification strategy is based on two assumptions. First, we assume that the number of households reporting incomes sufficiently far below the old notch is unimpacted by the reform (i.e., that frictions are limited). Under this assumption, the bunching effect and jumping effect can be estimated from the changes in bunching mass around the old notch and new notch. We then discretize the reported income distribution into bins and use the number of households reporting incomes in bins far below the original notch as controls for the number of households reporting incomes in bins around the old and new notches. We show that in the pre-reform period, the differences between the number of households in our treatment and control bins are well approximated by (low-order) polynomials. Our second identification assumption is that these differences between treatment and control bins would have persisted into the post-reform period in absence of the reform; we then employ a generalized difference-in-difference strategy (see, for example, Wolfers (2006) or Mora and Reggio (2013)). We perform numerous placebo exercises to support the validity of these two identification assumptions.

Using this generalized difference-in-difference strategy, we find that the number of households reporting incomes at and just below the new notch increased by approximately 49,000 and the number of households reporting incomes at and just below the old notch decreased by approximately 27,000 as a result of the reform. This translates into a lower bound for the MVPF of the reform of approximately 0.9 and an upper bound for the MVPF of approximately 1.12. In terms of welfare implications, as long as the government values giving R$0.90 (in a non-distortionary manner) to the BF households more than spending R$1 on their next best alternative, the reform was welfare improving. We argue that this is likely the case given that BF has high coverage of households in extreme poverty and given that the households misreporting to get the benefit typically fall in the poorest half of the population (Bastagli (2008) and Lindert et al. (2007)). A back-of-the-envelope calculation suggests that the welfare effect of spending R$1 on the BF reform is at least as high as the welfare effect of spending R$1.50 on a non-distortionary universal transfer. Hence, our empirical findings contribute to the debate around targeted vs. universal transfers in developing settings (Hanna and Olken, 2018): even in a setting with a highly pronounced notch and substantial scope for misreporting, the efficiency cost generated by behavioral responses is simply not large enough to outweigh the equity gain associated with the increased generosity of benefits targeted to poor households (relative to a universal transfer).

Relationship to the literature:First, we contribute to the large literature that explores bunching at notches and kinks. Typically papers in this literature use reduced-form bunching evidence to pin down structural parameters of interest (e.g., the elasticity of taxable income). These parameters can then be used to inform the welfare impacts of observed or proposed policy changes, although, in this literature, they are more commonly used to identify alternative schedules that would improve welfare (see, for example, Best et al. (2015) or Bachas and Soto (2021)). However, the chief limitation of this approach is that translating a bunching mass into a structural parameter typically relies heavily on the modeling assumptions of the household problem (Kleven, 2016). Hence, our approach is complementary to standard bunching approaches in the sense that they have opposing strengths and weaknesses: our framework avoids making strong assumptions on the household optimization problem, but our framework can only be used to analyze the impacts of observed reforms whereas the standard bunching approach recovers structural parameters that can be used to analyze hypothetical reforms or recover the optimal policy.

Furthermore, our empirical strategy differs from standard bunching analyses. The standard bunching approach uses portions of the distribution unimpacted by the notch combined with smoothness assumptions on the underlying counterfactual distribution to identify bunching induced by the notch. In contrast, we focus on estimating *changes* in bunching that result from a reform via a difference-in-difference strategy that uses portions of the distribution unimpacted by the reform to control for underlying time trends in portions of the distribution which are impacted by the reform.[4] As such, our approach has stronger data requirements: we require repeated cross-sectional data whereas standard bunching approaches as in Saez (2010) or Kleven and Waseem (2013) only require cross-sectional data.

Our paper also contributes to the sufficient statistics approach for welfare analysis (Chetty, 2009). One broad contribution to this literature is showing how to apply a sufficient statistics framework to settings with large, discrete policy reforms when the envelope condition cannot be applied. Kleven (2021) discusses how to do approximate welfare analysis using an expanded set of sufficient statistics when reforms are large; however, he argues that these additional statistics are difficult to estimate. We overcome the need to estimate a large set of complex statistics by focusing on welfare bounds. This paper is also related to Lockwood (2020) who investigates the welfare impacts of changing the top marginal tax rate in a piecewise linear tax system that features a notch, arguing that the sufficient statistic approach needs to be augmented to include a correction term which captures the change in bunching at the notch in response to a tax reform. Our key theoretical contribution relative to Lockwood (2020) is the generality of our results. Whereas Lockwood (2020) characterizes the welfare impacts of particular infinitesimal reforms to notches in a model with quasi-linear utility, one dimension of agent heterogeneity, and one choice variable, our approach allows us to bound the welfare impacts of reforms to programs with notches of any size while placing very little structure on the agent problem. Our framework also allows us to understand welfare impacts of changes in the location of notches, which turns out to be theoretically more complex.

Finally, we contribute to the small but growing literature on estimating the welfare impacts of cash transfer programs in developing countries (e.g., Bergolo and Cruces (2021); Bergstrom and Dodds (2021), Hanna and Olken (2018)). While we find strong evidence of behavioral responses to the BF notch reform, our sufficient statistics framework suggests

[4]Because we use unimpacted regions of the reported income distribution to form a counterfactual for the impacted region, our method is conceptually closer to the bunching literature than it is to papers such as Cengiz et al. (2019) who use the income distributions in unimpacted *geographical* areas to form counterfactuals for the income distribution in impacted *geographical* areas. We cannot apply this sort of methodology in our context because the BF reform was a national reform implemented uniformly across Brazil in June 2014.

that these responses are unlikely to be large enough to generate efficiency losses that outweigh the equity benefits.[5] Notably, because we analyze a reform to a program with a pronounced eligibility notch based on self-reported income in a high-informality setting, one may ex-ante expect the efficiency costs of the reform to be very large. The fact that we do not find sizable efficiency costs provides evidence against the commonly held belief that targeting transfer programs based on self-reported incomes will have substantial efficiency costs in high informality settings. Thus our findings may have implications for the future design of cash transfer programs in developing countries.

The remainder of the paper is organized as follows. Section I introduces the theoretical framework for welfare analysis of changing notches, Section II discusses the Bolsa Família program and our data, Section III discusses our strategy to empirically identify the bunching effect and jumping effect for the June 2014 reform, Section IV presents our results and robustness analysis, and Section V concludes.

## I. Theoretical Framework for Welfare Analysis of Changing Notches

In this section we develop a sufficient statistics framework to bound the welfare impacts of changing a notch in a transfer program. While we will apply this framework to the June 2014 BF reform, we believe this framework can be used more generally to analyze reforms to notches in other transfer programs. To begin, we derive bounds in a simple, static misreporting model. This simple setup is useful not only for building intuition but also because misreporting responses are likely common in response to the BF reform. We show that we can bound the welfare impact of a notch change using two empirical objects: (1) the jumping effect, which captures the change in the number of households locating at or below the new notch as a result of the reform and (2) the bunching effect, which captures the change in the number of households locating at or below the old notch as a result of the reform. We then show that these bounds hold in a much more general model that places minimal structure on the household problem, thereby arguing that our two empirical objects are "sufficient statistics" to bound the change in welfare from a notch change.

### A. *Baseline Model Set-Up*

To begin, we consider a static world in which households choose how much income to report, $\hat{y}$, subject to a policy $\mathbf{p} = \{b, \tau\}$ where $b$ denotes the level of the benefit and $\tau$ denotes the eligibility threshold s.t. those reporting $\hat{y} \leq \tau$ receive $b$ and those reporting $\hat{y} > \tau$ receive nothing. Households have two dimensions of heterogeneity: (1) endowed income $y$ distributed according to CDF $F(y)$, and (2) aversion to misreporting governed by $\mu \in \{1, 2\}$ with probability mass function $\pi(\mu)$. Type $\mu = 1$ households are "truth-telling"

[5] Similarly, Bergolo and Cruces (2021) estimate sizable equity benefits relative to efficiency costs for a cash transfer program in Uruguay that bases eligibility, in part, on reported incomes. They estimate that the MVPF is 0.61, which is in line with cash transfer programs in the United States for which Hendren and Sprung-Keyser (2020) estimate an average MVPF of 0.74.

households who never misreport, whereas type $\mu = 2$ households are willing to misreport their income. For simplicity, we assume there is no fixed cost of reporting an income, all households know the policy $\mathbf{p}$, and households have quasi-linear utility in consumption. Type $\mu = 1$ households therefore have utility under policy $\mathbf{p}$ given by:

$$U^*(y, \mu = 1; \mathbf{p}) \equiv y + b\mathbb{1}\left(y \leq \tau\right) \tag{1}$$

Utility for type $\mu = 2$ households under policy $\mathbf{p}$ is equal to:

$$\begin{aligned} U^*(y, \mu = 2; \mathbf{p}) \equiv \max_{\hat{y}} \;& c - v\left(y - \hat{y}\right) \\ \text{s.t. } & c = y + b\mathbb{1}\left(\hat{y} \leq \tau\right) \end{aligned} \tag{2}$$

where $v\left(y - \hat{y}\right)$ captures the disutility of reporting $\hat{y}$ when true income is $y$. We assume $v(y - \hat{y}) = 0$ if $\hat{y} \geq y$ (so that only under-reporting is costly) along with $v' > 0$ when $y > \hat{y}$ (so that the cost of misreporting is increasing in the discrepancy between true and reported income). Optimal reported income, $\hat{y}^*$, for a type $\mu = 2$ household equals (see Appendix A.1 for a formal derivation):

$$\hat{y}^*(y, \mu = 2; \mathbf{p}) = \begin{cases} y & \text{if } y \leq \tau \\ \tau & \text{if } y \in (\tau, y^c(\mathbf{p})] \\ y & \text{if } y > y^c(\mathbf{p}) \end{cases} \tag{3}$$

where $y^c(\mathbf{p})$ is the income level for which households are indifferent between misreporting at $\tau$ and reporting truthfully, implicitly defined by $y^c(\mathbf{p}) + b - v\left(y^c(\mathbf{p}) - \tau\right) = y^c(\mathbf{p})$. In words, type $\mu = 2$ individuals with $y < \tau$ report truthfully and get the benefit, those with $y \in (\tau, y^c(\mathbf{p})]$ get the benefit by misreporting and bunching at the notch, and those with $y > y^c(\mathbf{p})$ report their income truthfully and do not get the benefit.

Next, we define $G(z; \mathbf{p})$ as the number of households reporting an income less than or equal to $z$ under policy $\mathbf{p}$:

$$G(z; \mathbf{p}) = \sum_{\mu \in \{1,2\}} \int_Y \mathbb{1}(\hat{y}^*(y, \mu; \mathbf{p}) \leq z) dF(y|\mu)\pi(\mu)$$

Hence, $G(\tau; \mathbf{p})$ captures the number of households locating under the eligibility threshold $\tau$. Finally, we assume that social welfare under policy $\mathbf{p}$, $\mathcal{W}(\mathbf{p})$, is given by a weighted sum of household utilities less the budgetary cost of the policy multiplied by the shadow value of public funds $\lambda$, which captures the welfare gain of spending a dollar on the government's next best alternative. Hence, we assume that spending \$1 on the program comes at the cost of spending \$1 less on some other program which decreases welfare by a constant value

$\lambda$. Welfare under policy $\mathbf{p}$ is therefore given by:

$$\mathcal{W}(\mathbf{p}) = \sum_{\mu \in \{1,2\}} \int_Y \phi(y,\mu) U^*(y,\mu;\mathbf{p}) dF(y|\mu)\pi(\mu) - \lambda b G(\tau;\mathbf{p}) \tag{4}$$

where $\phi(y,\mu)$ denotes the government's welfare weight on a household with income $y$ and type $\mu$.

### B. *Welfare Effect of the Reform in the Baseline Model*

Our goal is to evaluate the welfare impact of a reform from policy $\mathbf{p} = \{b,\tau\}$ to $\mathbf{p}' = \{b',\tau'\}$, defining $\{\Delta b, \Delta\tau\} = \mathbf{p}' - \mathbf{p}$. For ease of exposition, let us assume $\Delta b > 0$ and $\Delta\tau > 0$ (i.e., we increase the level and location of the notch). Note that we are *not* restricting to infinitesimal reforms; the bounds we derive allow for arbitrary, discrete reforms to $b$ and $\tau$. We derive welfare bounds for infinitesimal reforms in Appendix A.4: the lower bound is identical as in the non-infinitesimal case but the upper bound can be tightened in the infinitesimal case (although this tighter bound is more difficult to estimate). Our goal is to derive bounds for $\mathcal{W}(\mathbf{p}') - \mathcal{W}(\mathbf{p})$ in terms of empirically observable objects (we discuss the advantages of focusing on bounding the welfare impact as opposed to characterizing the exact impact in Section I.C). Hendren and Sprung-Keyser (2020) show that the welfare impacts of any policy reform can be expressed in terms of a *normative* welfare weight along with a *positive* sufficient statistic called the marginal value of public funds (MVPF), which captures households' willingness-to-pay (WTP) for the reform relative to the total budgetary cost of the reform. Our goal then is to derive bounds for the MVPF.

Because the utility function is quasi-linear, a household with income $y$ and misreporting type $\mu$ has a WTP for the reform implicitly defined by:

$$U^*(y,\mu;\mathbf{p}) = U^*(y,\mu;\mathbf{p}') - WTP$$

Equivalently, WTP is equal to the compensating variation. We now heuristically derive bounds on the WTP for all households impacted by the reform. We split the set of households impacted by the reform into four groups: the mechanical households, the bunching households, the threshold households, and the jumping households. Figure 1 depicts these four groups for hypothetical reported income distributions under policies $\mathbf{p}$ and $\mathbf{p}'$.

The mechanical households are the households who report at or below $\tau$ under policy $\mathbf{p}$ and who do not change their behavior in response to the reform. Because the reform increases benefits for mechanical households by $\Delta b$, their WTP for the reform is simply $\Delta b$. The number of mechanical households is given by $M = G(\tau;\mathbf{p}')$ (notably, $M$ is not equal to the number of households reporting at or below $\tau$ under policy $\mathbf{p}$ as this includes bunching households who do change their behavior as a result of the reform).

The bunching households are the households who misreport and bunch at the original threshold $\tau$ under policy $\mathbf{p}$; these households move with the threshold as it is increased

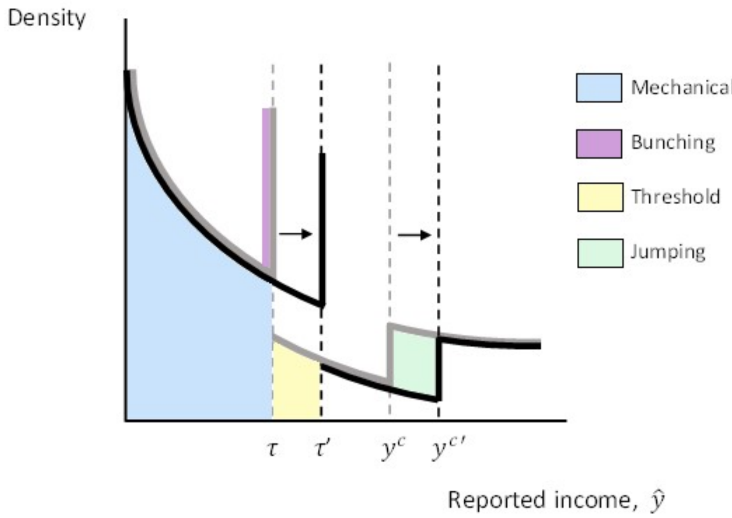



Figure 1. A Hypothetical Density of Reported Incomes under $\mathbf{p}$ and $\mathbf{p}'$

*Note:* This figure shows a hypothetical density of reported incomes under the initial policy $\mathbf{p} = \{b, \tau\}$ (solid grey curve) and how this density changes as a result of a reform that increases the policy to $\mathbf{p}' = \{b', \tau'\}$ (solid black curve). Note, the vertical solid grey line at $\tau$ and the vertical solid black line at $\tau'$ represent the bunching households under the initial policy and new policy, respectively. This figure also depicts which households are classified as mechanical households, bunching households, threshold households, and jumping households.

(i.e., they report between $(\tau, \tau']$ under policy $\mathbf{p}'$).[6] Bunching households receive an increase in benefits equal to $\Delta b$. Moreover, they experience a reduction in their misreporting costs as they move from $\tau$ to $\tau'$.[7] Hence, these households have a WTP of at least $\Delta b$. Moreover, by revealed preference arguments, the most the bunching households can value this reduction in misreporting costs is $b$ dollars. If they value this reduction by more than $b$ dollars, it would not have been optimal for these households to bunch at $\tau$ under policy $\mathbf{p}$.[8] Thus, the WTP of bunching households is in $[\Delta b, \Delta b + b] = [\Delta b, b']$. The number of bunching households is equal to what we term the "bunching effect", which is defined as the causal impact of the reform on the number of households locating at or below the original threshold $\tau$: $B = G(\tau; \mathbf{p}) - G(\tau; \mathbf{p}')$. It will eventually be important to

[6]Not all bunching households will move all the way to $\tau'$: bunchers with $y \in (\tau, \tau')$ will report $\hat{y} = y$ under $\mathbf{p}'$.

[7]Even for an infinitesimal reform, the utility gain that bunching households experience from moving with the notch has a first-order impact on welfare. This is because the envelope theorem cannot be applied for these individuals. In order to apply the envelope theorem, utility must be differentiable with respect to the policy given *any* fixed choices (see Theorem 2 of Milgrom and Segal (2002)). However, utility is actually discontinuous as a function of the parameter $\tau$ holding decisions fixed: for example, individuals reporting an income of $\tau$ see a discrete drop in consumption (and hence utility) if $\tau$ is reduced by any amount. See Appendix A.4 for more details.

[8]Suppose bunching households have a WTP for the relaxation in their misreporting costs greater than $b$: $v(y - \tau) - v(y - \tau') > b$. This implies $v(y - \tau) > b$. However if $v(y - \tau) > b$, these households would have preferred to report truthfully above $\tau$ over misreporting at $\tau$ under policy $\mathbf{p}$.

distinguish between the "bunching effect" $B$ (the causal impact of the reform on the number of households locating at or below the original threshold) and the "number of bunching households" (the number of households who misreport and bunch *precisely* at the original threshold). While the bunching effect $B$ is equal to the number of bunching households in the present misreporting model, this will no longer be true in the more general model that we will discuss in Section I.C due to, for example, households that cannot precisely bunch at the initial threshold as a result of frictions yet still move beyond the old notch towards the new notch as a result of the reform. Previewing ahead, the bunching effect $B$ will nonetheless continue to be one of the two sufficient statistics needed to bound the welfare impact of a notch reform even in this more general model.

The threshold households are the households who report in $(\tau, \tau']$ under policy $\mathbf{p}$. These households will not change their behavior in response to the reform because all households reporting above $\tau$ given policy $\mathbf{p}$ are reporting truthfully regardless of their type $\mu$. Under $\mathbf{p}'$ the optimal choice for these households is to continue reporting truthfully as doing so allows them to receive $b'$ and not incur any misreporting costs. These households go from receiving no benefits to receiving $b'$ in benefits. Hence their WTP is equal to $b'$. The number of threshold households is given by $T = G(\tau'; \mathbf{p}) - G(\tau; \mathbf{p})$.

Finally, the jumping households are the households who report above $\tau'$ given policy $\mathbf{p}$ but who report at $\tau'$ given policy $\mathbf{p}'$. In particular, households who were previously close-to-indifferent between misreporting at the threshold and truthfully reporting above the threshold but opted for the latter, i.e., type $\mu = 2$ households with $y \in (y^c(\mathbf{p}), y^c(\mathbf{p}')]$, will now jump and misreport to the new threshold $\tau'$.[9] We describe the behavioral response of these households as a "jump" because these households change their optimal reported income discontinuously as we move from $\mathbf{p}$ to $\mathbf{p}'$. By revealed preference, jumping households' utility is improved by changing their behavior; hence, their WTP is weakly positive. Moreover, after the reform, jumping households get $b'$ more dollars but incur misreporting costs. Therefore jumping households' WTP cannot exceed $b'$ as misreporting costs are weakly positive. Hence the WTP of jumping households is in $[0, b']$. The number of jumping households is equal to what we term the "jumping effect", which is defined as the causal impact of the reform on the number of households locating at or below the new threshold $\tau'$: $J = G(\tau'; \mathbf{p}') - G(\tau'; \mathbf{p})$. As with the bunching effect vs. number of bunching households, it is important to distinguish between the "jumping effect" (the causal impact of the reform on the number of households locating at or below the new threshold) and the "number of jumping households" (the number of households who jump down to bunch *precisely* at the new threshold due to the reform). While the jumping effect equals the number of jumping households in the present misreporting model, in our more general model in Section I.C the jumping effect may also comprise households that jump down to income levels strictly below the new threshold due to frictions; nonetheless, the jumping effect $J$ will continue to be one of the two sufficient statistics to bound the welfare impact of a notch reform.

[9]For changes in $\tau$ s.t. $y^c(\mathbf{p}) < \tau'$, only $\mu = 2$ households with $y \in (\tau', y^c(\mathbf{p}')]$ jump and misreport to the new notch. See Appendix A.5 for Figure 1 under this scenario.

Table 1 summarizes the bounds on the WTP for each of our four groups along with the number of households falling into each group and the cost that each group imposes on the government.

Table 1— WTP and Cost to the Government of the Reform

| Group | Number of households | Utility under $\mathbf{p}$ (per household) | Utility under $\mathbf{p}'$ (per household) | WTP (per household) | Cost to Govt. (per household) |
|---|---|---|---|---|---|
| Mechanical | $M = G(\tau; \mathbf{p}')$ | $y + b$ | $y + b'$ | $\Delta b$ | $\Delta b$ |
| Bunching | $B = G(\tau; \mathbf{p}) - G(\tau; \mathbf{p}')$ | $y + b - v(y - \tau)$ | $y + b' - v(y - \hat{y}^*(\mathbf{p}'))$ | $\in [\Delta b, b']$ | $\Delta b$ |
| Threshold | $T = G(\tau'; \mathbf{p}) - G(\tau; \mathbf{p})$ | $y$ | $y + b'$ | $b'$ | $b'$ |
| Jumping | $J = G(\tau'; \mathbf{p}') - G(\tau'; \mathbf{p})$ | $y$ | $y + b' - v(y - \tau')$ | $\in [0, b']$ | $b'$ |

*Note:* This table shows utility under both policy regimes $\mathbf{p}$ and $\mathbf{p}'$ along with willingness-to-pay (WTP) and the cost to the government for all the households impacted by the reform from policy $\mathbf{p}$ to $\mathbf{p}'$. Note, for bunching households, $\hat{y}^*(\mathbf{p}') \in (\tau, \tau']$.

This brings us to Lemma 1 which bounds the total WTP for the reform:

**Lemma 1.** *If individuals solve Problem 2, the total WTP of the reform from* $\mathbf{p}$ *to* $\mathbf{p}'$ *with* $\mathbf{p}' - \mathbf{p} = \{\Delta b, \Delta \tau\} > 0$ *can be bounded as follows:*

$$\Delta b(M + B) + b'T \leq \textit{Total WTP} \leq \Delta b M + b'(B + T + J)$$

*where* $M, B, T$ *and* $J$ *are defined in Table 1.*

*Proof.* See Appendix A.2. □

Moreover, using the cost per-household to the government given in Table 1, we can express the total budgetary cost of the reform as follows:

$$\text{Total Cost} \equiv b'G(\tau'; \mathbf{p}') - bG(\tau; \mathbf{p}) = \Delta b(M + B) + b'(T + J) \tag{5}$$

Hence, we can construct bounds for the MVPF of the reform:

**Proposition 1.** *If individuals solve Problem 2, the marginal value of public funds of the reform from* $\mathbf{p}$ *to* $\mathbf{p}'$ *with* $\mathbf{p}' - \mathbf{p} = \{\Delta b, \Delta \tau\} > 0$ *can be bounded as follows:*

$$MVPF \geq MVPF_L \equiv \frac{\Delta b(M + B) + b'T}{\Delta b(M + B) + b'(T + J)} = 1 - b' \frac{J}{\textit{Total Cost}}$$

$$MVPF \leq MVPF_U \equiv \frac{\Delta b M + b'(B + T + J)}{\Delta b(M + B) + b'(T + J)} = 1 + b \frac{B}{\textit{Total Cost}}$$

*Proof.* This follows directly from Lemma 1 and Equation 5. □

The lower bound for the MVPF captures the fact that the lower bound for the WTP of the jumping households is 0 while the cost they impose on the government is $b'$ each (whereas the lower bound for the WTP of bunching households is exactly equal to the cost they impose on the government). Hence, if all jumping households value the reform at their lower bound, $b'J/(\text{Total Cost})$ of each dollar of spending is "wasted". Meanwhile, the upper bound for the MVPF captures the fact that the upper bound for the WTP of the bunching households is $b'$ while the cost they impose on the government is only $b' - b$ (whereas the upper bound for the WTP of jumping households is exactly equal to the cost they impose on the government). Hence, if bunching households all value the reform at their upper bound, the government "gains" $bB/(\text{Total Cost})$ for each dollar of spending. Because $B, J > 0$, the MVPF bounds in Proposition 1 will always contain 1. Intuitively, the bounds in Proposition 1 will be tighter when the number of mechanical and threshold households is large relative to the number of jumping and bunching households. We can then use these MVPF bounds to construct bounds on the money metric welfare gain relative to the budgetary cost of the reform using Proposition 2:[10]

**Proposition 2.** *If individuals solve Problem 2 and social welfare is given by Equation 4, then the money metric welfare gain relative to the budgetary cost of the reform from* $\mathbf{p}$ *to* $\mathbf{p}'$ *with* $\mathbf{p}' - \mathbf{p} = \{\Delta b, \Delta\tau\} > 0$ *can be bounded as follows:*

$$\omega_L MVPF_L - 1 \leq \frac{\frac{1}{\lambda}\left[\mathcal{W}(\mathbf{p}') - \mathcal{W}(\mathbf{p})\right]}{b'G(\tau'; \mathbf{p}') - bG(\tau; \mathbf{p})} \leq \omega_U MVPF_U - 1$$

*where* $\omega_L$ *(*$\omega_U$*) denotes the weighted average money-metric welfare gain from giving a dollar to mechanical, bunching, threshold, and jumping households, where the weights are determined by the relative size of each group's lower bound (upper bound) WTP for the reform.*

*Proof.* See proof of Proposition 3 (which nests Proposition 2) in Appendix A.3. □

Proposition 2 bounds the increase in total welfare from spending \$1 on the reform. In particular, $MVPF_L$ captures our lower bound on the total WTP when we spend \$1 on the reform, while $\omega_L$ denotes the incidence-weighted welfare gain, measured in dollars, of splitting \$1 among the mechanical, bunching, threshold, and jumping households. In other words, $\omega_L$ is the average welfare weight for mechanical households multiplied by the (lower bound) share of WTP that goes to mechanical households plus the average welfare weight for threshold households multiplied by the (lower bound) share of WTP that goes to threshold households, etc.[11] Subtracting the budgetary cost of \$1 from $\omega_L MVPF_L$ gives a lower bound for the total welfare gain from spending \$1 on the reform. Symmetric

[10]Note, that the welfare change is expressed in dollar units (as opposed to welfare units) as we divide through by the shadow value of public funds, $\lambda$.

[11]These incidence-weighted welfare weights are identical to those used in Hendren and Sprung-Keyser (2020). We provide a more detailed example in Appendix A.6 showing how these incidence-weighted welfare weights interact with the MVPF to make welfare statements.

logic explains why $\omega_U MVPF_U - 1$ is an upper bound for the increase in total welfare of spending \$1 on the reform. Intuitively, these welfare bounds will be informative when either the lower bound is close to 1 and the government's welfare weight on the affected group of households is high (in this case the policy reform is welfare increasing) or when the upper bound is close to 1 and the government's welfare weight on the affected group of households is low (in this case the policy reform is welfare decreasing). Because Proposition 1 tells us that $MVPF_L$ and $MVPF_U$ are functions of two objects, the bunching effect and the jumping effect, Proposition 2 implies that the bunching effect and the jumping effect are the empirical objects needed to construct bounds for the welfare impact of the reform.[12]

### C. Robustness to Model Specification

To highlight the robustness of our welfare bounds, we now show that we actually require very little structure on preferences or behavioral responses of households. Suppose households have several decisions variables denoted by the vector $\mathbf{x}$ within a choice set $\mathbf{X}$. Household decisions are made conditional on primitives denoted by the vector $\theta \in \Theta$ and the policy $\mathbf{p}$. Households get the benefit $b$ if their reported income $\hat{y}$, which is a function of decision variables and primitives, is below $\tau$. Household income, denoted $y$, is also potentially a function of decision variables. For example, we could consider a model where households have two working adults, $\mathbf{x}$ is a vector representing two labor supply decisions $l_1$ and $l_2$ along with a choice of reported income $\hat{y}$, $\theta$ is a vector representing two productivities $n_1$ and $n_2$ along with an aversion to misreporting $\mu$, and household income is given by $y = n_1 l_1 + n_2 l_2$. In such a model households can respond to reforms via changing their labor supply as well as misreporting. More generally, we consider the following household problem:

$$
\begin{aligned}
U^*(\theta; \mathbf{p}) = &\max_{\mathbf{x} \in \mathbf{X}} \; u\left(c, \mathbf{x}; \theta\right) \\
&\text{s.t. } \; c = y(\mathbf{x}, \theta) + b\mathbb{1}(\hat{y}(\mathbf{x}, \theta) \leq \tau)
\end{aligned} \tag{6}
$$

where $c$ denotes consumption. We assume total welfare is given by a weighted sum of utilities, with welfare weights given by $\phi(\theta)$:

$$
\mathcal{W}(\mathbf{p}) = \int_{\Theta} \phi(\theta) U^*(\theta; \mathbf{p}) dF(\theta) - \lambda b G(\tau; \mathbf{p}) \tag{7}
$$

where $\lambda$ represents the shadow value of public funds and $G(\tau; \mathbf{p}) = \int_{\theta: \hat{y}(\theta, \mathbf{p}) \leq \tau} dF(\theta)$ represents the number of households receiving the benefit under policy $\mathbf{p}$. More generally, we define $G(z; \mathbf{p}) = \int_{\theta: \hat{y}(\theta, \mathbf{p}) \leq z} dF(\theta)$. This setup allows us to substantially generalize Proposition 1 and Proposition 2:

[12] Technically, we also need to estimate the total cost of the reform; however, Equation 5 shows that the total cost is a function solely of $G(\tau'; \mathbf{p}')$ and $G(\tau; \mathbf{p})$, which are components needed to construct $B$ and $J$.

**Proposition 3.** *Suppose households solve Problem 6 and welfare is given by Equation 7. Define:*

$$MVPF_L \equiv \frac{\Delta b G(\tau;\mathbf{p}) + b'[G(\tau';\mathbf{p}) - G(\tau,\mathbf{p})]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p})} = 1 - \frac{b'[G(\tau';\mathbf{p}') - G(\tau';\mathbf{p})]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p})} = 1 - b'\frac{J}{Total\ Cost} \tag{8}$$

$$MVPF_U \equiv \frac{\Delta b G(\tau;\mathbf{p}') + b'[G(\tau';\mathbf{p}') - G(\tau,\mathbf{p}')]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p})} = 1 + \frac{b[G(\tau;\mathbf{p}) - G(\tau;\mathbf{p}')]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p})} = 1 + b\frac{B}{Total\ Cost} \tag{9}$$

*Then as long as $\tau' > \tau$ and $b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p}) > 0$ we have:*

$$\omega_L MVPF_L - 1 \leq \frac{\frac{1}{\lambda}[\mathcal{W}(\mathbf{p}') - \mathcal{W}(\mathbf{p})]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p})} \leq \omega_U MVPF_U - 1 \tag{10}$$

*where $\omega_L$ ($\omega_U$) denotes the weighted average money-metric welfare gain from giving a dollar to households impacted by the reform where the weights are determined by the relative size of each households's lower bound (upper bound) WTP for the reform.*

*Proof.* See Appendix A.3. □

Proposition 3 highlights that we can bound the MVPF and, consequently, bound the welfare impacts of the reform using empirically observable objects while only putting limited structure on the household problem.[13] In particular, to bound the MVPF we just need to estimate the jumping effect $J = G(\tau';\mathbf{p}') - G(\tau';\mathbf{p})$, which captures how the number of people locating at or below the new notch changes as a result of the reform, and the bunching effect $B = G(\tau;\mathbf{p}) - G(\tau;\mathbf{p}')$, which captures how the number of people locating at or below the old notch changes as a result of the reform. Recall that in the context of our simple baseline model, $J$ equals the number of households jumping down to bunch *precisely* at the new notch and $B$ equals the number of households who bunch *precisely* at the original notch. However, in the more general framework of Proposition 3, the jumping effect comprises not only households that jump down to bunch exactly at the new threshold but also households that jump down *anywhere* below the new notch and the bunching effect comprises not only the reduction in bunching exactly at the old notch but also any other change in the number of households anywhere below the old notch. To see why this is the case, consider a labor supply model with a limited choice set (a type of friction) in which households can only choose discrete incomes in $\{y_1, y_2, y_3, ...\} \not\ni \tau, \tau'$. In such a setting, households cannot precisely bunch at either the old or new notch; hence, $J = G(\tau';\mathbf{p}') - G(\tau';\mathbf{p})$ just captures the total number of households that jump down to

[13]Assuming $\tau' > \tau$ in Proposition 3 is WLOG - we are just arbitrarily labeling policy $\mathbf{p}'$ as the one with larger $\tau'$. Moreover, in Proposition 3 we also assume that the policy reform leads to a net budgetary cost so that $b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p}) > 0$; this is also WLOG as if the policy reform leads to a net budgetary revenue gain, then both inequalities in Equation 10 are reversed.

*any* income $y_i \leq \tau'$ as a result of the reform. Similarly, $B = G(\tau; \mathbf{p}) - G(\tau; \mathbf{p}')$ captures any change in the number of households anywhere below the old notch $\tau$ as a result of the reform (e.g., households moving from $y_i \leq \tau$ to $y_i \in (\tau, \tau'])$. Intuitively, in a world with frictions, the jumping effect and bunching effect must account for changes strictly below the new and old notches due to diffuse bunching.

Problem 6 can encompass a variety of important realisms: (1) households may respond by changing labor supply instead of misreporting their income (or respond on a variety of dimensions), (2) households may face limited choice sets (e.g., restrictions on labor supply), (3) households may face reporting costs (e.g., hassle or time costs) and thereby face a decision of whether to report/update their income on the registry[14], (4) households may have a wide range of heterogeneity in their utility functions (e.g., households may have varying preferences over the labor/leisure trade-off or varying preferences to locate at round numbers). The intuition for why Proposition 3 is so general comes from revealed preference arguments. We derive a lower bound for the MVPF of a policy shift from $\mathbf{p}$ to $\mathbf{p}'$ by calculating how welfare would change if no one re-optimized (by revealed preference, behavioral responses can only improve welfare). Similarly we derive an upper bound for the MVPF of a policy shift from $\mathbf{p}'$ to $\mathbf{p}$ by starting from policy $\mathbf{p}'$ and thinking about how welfare would change as we move to policy $\mathbf{p}$ if no one re-optimized from their choices under $\mathbf{p}'$. Hence, our bounds are robust to the types of behavioral responses available to households along with the restrictions on household choice sets precisely because behavioral responses are held fixed when formulating our MVPF bounds. We view this robustness as perhaps the most important aspect of our theory.

But of course there are *some* implicit restrictions encoded in the assumed household problem (Problem 6) used to prove Proposition 3. Perhaps most importantly, Proposition 3 requires that households correctly perceive the benefit schedule and the reform. The proof to Proposition 3 uses the fact that household re-optimization *improves* utility; however, Proposition 3 is robust to household misperceptions as long as, on average, behavioral responses to reforms improve welfare (i.e., misperceptions are not so large that households, on average, harm themselves by responding to reforms; see Appendix A.7). For example, if some proportion of households are entirely unaware of the reform while the rest of the population is perfectly aware of the reform, our bounds hold as unaware households will not respond to the reform whereas those who are aware of the reform improve their utility via their behavioral response.

Moreover, Problem 6 is also a static problem. We augment Proposition 3 in Appendix A.8 to show that we can bound the discounted welfare impact of the policy over time relative to the discounted total budgetary cost in a general dynamic model allowing for income dynamics, savings, and stochastic shocks. In this case, the relevant bounds for the MVPF are constructed using the discounted sum of the expected jumping effect over time and the discounted sum of the expected number bunching effect over time.

[14]To capture adjustment/updating costs, one could suppose $\mathbf{x} = \hat{y}_t$ and $\theta$ consists of current income $y_t$, aversion to misreporting $\mu$, and prior reported income $\hat{y}_{t-1}$; households incur an adjustment cost $k$ if their reported income today differs from prior reported income.

Additionally, Proposition 3 can be augmented to allow for more complex policy environments. For instance, Proposition 3 holds even if households only receive the benefit with some probability if they report below the notch (see Appendix A.9). The intuition being that this probability appears in both the numerator and denominator of the MVPF bounds (as it impacts both households' WTP for the reform as well as the total cost of the reform) and thus cancels out.[15] Moreover, while Proposition 3 assumes that there is no underlying tax and transfer system beyond the benefit $b$ given to those with a reported income $\hat{y} \leq \tau$, Proposition 3 can be augmented to provide welfare bounds of a policy reform to the benefit and threshold level of a notch when there is a more complex underlying tax and transfer schedule; in this case, the total budgetary cost of the reform must include the impacts that behavioral responses have on other programs that impact the government's budget (i.e., we need to calculate the additional fiscal externalities associated with the reform).[16]

Lastly, we discuss the advantages of bounding the welfare impacts (as opposed to exactly characterizing the welfare impacts) of a notch reform. First, because we cannot appeal to envelope conditions in our setting, exactly characterizing welfare impacts requires one to take a stance on which margins households are responding, the functional form of their utility function, the sorts of frictions they face, etc. In contrast, bounding welfare impacts requires very little structure on the household optimization problem. In fact, Appendix A.10 provides examples wherein the upper and lower bounds in Proposition 3 are attained, thereby proving that the bounds in Proposition 3 cannot be tightened without making additional assumptions on primitives. Second, our bounds on the welfare impact of a notch change are expressed in terms of estimable reduced-form objects, whereas exactly characterizing welfare impacts would require one to estimate a potentially large number of structural parameters.

[15] If the probabilities of receiving the benefit vary by groups, the MVPF bounds must be augmented to account for differing probabilities (we omit a proof for brevity, the logic is similar to Appendix A.9). Letting $q_M, q_B, q_T, q_J$ denote the average probability of receiving the benefit for mechanical, bunching, threshold, and jumping households we have:

$$MVPF \geq MVPF_L \equiv \frac{q_M \Delta bM + q_B \Delta bB + q_T b'T}{q_M \Delta bM + q_B \Delta bB + q_T b'T + q_J b'J} = 1 - q_J b' \frac{J}{\text{Total Cost}}$$

$$MVPF \leq MVPF_U \equiv \frac{q_M \Delta bM + q_B b'B + q_T b'T + q_J b'J}{q_M \Delta bM + q_B \Delta bB + q_T b'T + q_J b'J} = 1 + q_B b \frac{B}{\text{Total Cost}}$$

[16] In particular, the total cost of the reform would equal $b'G(\tau';\mathbf{p}') + R(\mathbf{p}') - bG(\tau;\mathbf{p}) - R(\mathbf{p})$ where $R(\mathbf{p})$ denotes net government spending under policy $\mathbf{p}$ (exclusive of spending on BF), and $\Delta R = R(\mathbf{p}') - R(\mathbf{p})$ captures the additional fiscal externalities of the reform. In this case, the lower and upper bounds for the MVPF are given by:

$$MVPF_L \equiv \frac{\Delta bG(\tau;\mathbf{p}) + b'[G(\tau';\mathbf{p}) - G(\tau,\mathbf{p})]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p}) + \Delta R} = 1 - b' \frac{J}{\text{Total Cost}} - \frac{\Delta R}{\text{Total Cost}}$$

$$MVPF_U \equiv \frac{\Delta bG(\tau;\mathbf{p}') + b'[G(\tau';\mathbf{p}') - G(\tau,\mathbf{p}')]}{b'G(\tau';\mathbf{p}') - bG(\tau;\mathbf{p}) + \Delta R} = 1 + b \frac{B}{\text{Total Cost}} - \frac{\Delta R}{\text{Total Cost}}$$

These bounds follow from identical logic as in Appendix A.3 if the cost of a given policy is $bG(\tau;\mathbf{p}) + R(\mathbf{p})$ and the household budget constraint is given by $c = y(\mathbf{x},\theta) + b\mathbb{1}(\hat{y}(\mathbf{x},\theta) \leq \tau) + T(\mathbf{x},\theta)$ where $T(\mathbf{x},\theta)$ denotes net transfers from the government exclusive of the benefit $b$. Thus, in addition to $B$ and $J$, we need to observe $\Delta R$ to assess the welfare impact of the reform.

## II. The Bolsa Família Program

Next, we turn to our empirical application: bounding the MVPF of a notch reform to the Brazilian Bolsa Família program. This section will discuss the Bolsa Família program, the June 2014 reform, as well as our data; Section III will dicuss how we identify the sufficient statistics discussed in Section I.

### *A. Program Details*

The Brazilian anti-poverty program Bolsa Família (BF), which has been recently supplanted by Auxílio Brasil, provided cash transfers to poor households based on their reported, monthly per-capita income. BF was implemented in October 2003 and was administered by the social development ministry (Ministério do Desenvolvimento Social, or MDS). BF was one of the world's largest cash transfer programs with around 14 million households receiving benefits in 2014 (Gazola Hellmann, 2015). Applicants reported their information, including household income, expenditures, assets, socioeconomic characteristics, and demographic characteristics to interviewers at Cadastro Único agencies, which were program offices spread across Brazil's 5,570 municipalities. Beneficiaries of the BF program were required to update their information once every 2 years to maintain their benefits.

Eligibility for the BF program was based on reported household per-capita income. Household per-capita income was calculated in five steps. First, the applicant was asked to report labor income for the last month as well as average monthly labor income over the past year for each member in the household (the applicant must present a government-issued ID for herself and for each family member thus making it difficult to register fictitious family members). Second, the applicant was asked to report average monthly income received from five additional sources for each household member (see Appendix B.1, which shows the questionnaire used to calculate monthly income). Third, for each individual, a computer calculated the minimum between the average monthly labor income over the past year and last month's labor income. Fourth, a computer summed this minimum monthly labor income along with income from the five additional sources to get a measure of total monthly income for each individual. Finally, a computer summed this individual total monthly income across all household members and divided it by the number of household members.

Households were classified as eligible if their reported per-capita income fell below one of two thresholds. First, households reporting a per-capita income below the "extreme-poverty threshold" were eligible for an unconditional benefit (referred to as the "basic benefit"). Second, households with children reporting a per-capita income below the higher "poverty threshold" were eligible for a conditional benefit (referred to as the "variable benefit") provided that they made health and education investments in their children. Notably, not all households reporting incomes below these thresholds received the benefit. This is because there was a quota (cap) on the number of beneficiaries per municipality. Prior to 2009, these quotas were based on the predicted number of households below the poverty threshold in each municipality. Post 2009, these quotas were based on the predicted number of households below the poverty threshold scaled by 1.18 (Gerard, Naritomi and Silva, 2021). Hence, if quotas were binding in a given municipality, some eligible households may not have received benefits.[17]

Finally, the MDS had several enforcement mechanisms to prevent income misreporting. First, during the interview, the income questions came at the end of the questionnaire so that questions on expenditures and assets could help the interviewer asses the veracity of reported income (Bastagli, 2008). Second, during the interview, the applicant was reminded of her responsibility to provide true statements under penalty of losing the right to be eligible for government programs (Gazola Hellmann, 2015). Third, the ministry conducted audits, which could be triggered by citizens' complaints and cross-checks of registry data with other datasets such as administrative data on formal employment, deaths, or automobile purchases (Gazola Hellmann, 2015). However, despite these attempts, the large informal sector in the Brazilian economy generated substantial scope for income misreporting.

### *B. The Transfer Schedule and the June 2014 Reform*

Between July 2009 and June 2014, the extreme-poverty threshold was R\$70 per-capita, per-month and the poverty threshold was R\$140 per-capita, per-month.[18] The basic (unconditional) benefit was equal to R\$70 per-month, while the variable (conditional) benefits were based on the number and ages of the children in the household (see Appendix B.2 for more information on the variable benefits). In June 2014, the government increased both the benefits and thresholds by 10%. Thus, the extreme-poverty threshold was raised from R\$70 to R\$77 per-capita, per-month, the poverty threshold was raised from R\$140 to R\$154 per-capita, per-month, the basic benefit was raised from R\$70 to R\$77 per-month, and the variable benefits were also increased by 10%. This reform was announced on national television by the president in April 2014.

Our main empirical analysis will focus on single individual households (households with one adult and no children). These households are not eligible for the variable benefits as they do not have children. Thus, prior to June 2014, these households were eligible for R\$70 per-month if their reported income was less than or equal to R\$70 per-month and 0 otherwise. After June 2014, these households were eligible for R\$77 per-month if their reported income was less than or equal to R\$77 per-month and 0 otherwise. Thus, the benefit schedule for these households has a single notch which increased both in level and location as a result of the June 2014 reform; see Figure 2.

[17] Recall that in Section I.C we discussed how our framework can be augmented when only a fraction of eligible households receive benefits. In Section IV.B, we will discuss why this feature of the program is unlikely to impact our MVPF bounds.

[18] In 2003, the extreme-poverty threshold and the poverty threshold were set to equal one-fourth and one-half of the monthly minimum wage of R\$200, respectively. These thresholds were periodically adjusted for inflation; the time betwen adjustments was *ad hoc* and not linked to the minimum wage. Prior to June 2014, the last readjustment was in July 2009 (Gazola Hellmann, 2015).

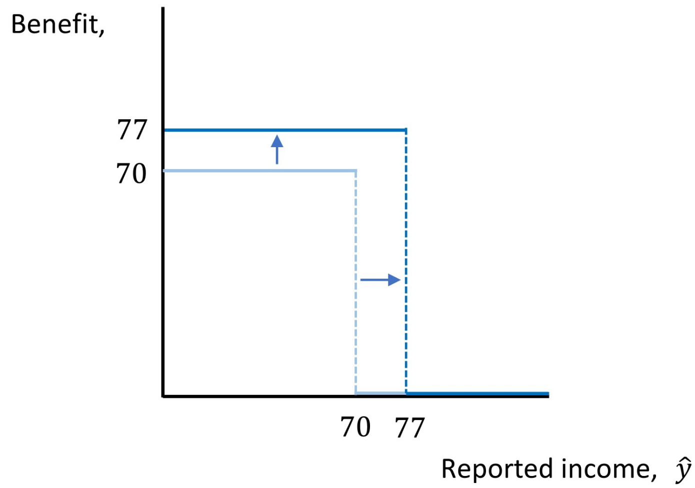



Figure 2. June 2014 Reform for Single Individual Households

We focus on single individual households because in February 2013, the government instituted a guaranteed minimum income of R$70 per-capita for all households which was subsequently raised to R$77 in the June 2014 reform. However, because the basic benefit is equal to the guaranteed minimum income, the benefit schedule for single individual households is not impacted by this minimum. In contrast, for all other households, this minimum creates a kink in the benefit schedule below the extreme-poverty threshold (the location of this kink will vary based on household composition); moreover, the location of this kink changed with the 2014 reform. For example, prior to the reform, households with two adults and no children had a kink at the reported per-capita income level of R$35 which increased to R$38.5 post June 2014.[19] This is problematic because, as will be discussed in Section III, one of our identification assumptions is that the reported income distribution sufficiently far below the extreme-poverty threshold is unaffected by the June 2014 reform; this is potentially less likely to hold for households with more than one member because the kink created by the guaranteed minimum income changes concurrently with the notch. Nonetheless, we will discuss the impacts of the June 2014 reform on households with more than one individual in Section IV.E.

Finally, in June 2016, there was another reform to the BF program that further increased both the benefit and the threshold. This reform, like the June 2014 reform, affected households of all compositions. As discussed next, our data ends in September 2016; thus, we do not have sufficient data beyond June 2016 to analyze the June 2016 reform.

[19] Prior to June 2014, a two adult household with a reported per-capita income less than R$35, $\hat{y} < 35$, will receive an additional monthly benefit equal to $2(70 - \hat{y}) - 70$. E.g., a two member household reporting a per-capita income of R$20 will receive R$70 in the basic benefit and an additional benefit of R$30.

### C. Data Sources and Sample Description

We have access to the Cadastro Único household registry, which is used to determine the eligibility of households for BF as well as all other targeted federal social programs (Veras Soares, 2011). Many of these other programs have eligibility criteria above the BF thresholds which explains the large number of ineligible applicants in the registry (see Table 2).[20] These other programs do not change concurrently with the BF reform that we analyze and are discussed in more detail in Appendix B.3.

Our final dataset is constructed by appending eight extractions of the registry: one in December of each year from 2011 until 2015, one in April 2015, one in August 2015, and one in September 2016.[21] Each extraction contains the latest information for all households on the registry at the time of the extraction date. For instance, if a household updated its information in August 2011 and September 2013, its information will appear as of August 2011 in the 2011 and 2012 extractions and as of September 2013 in the 2013, 2014, 2015, and 2016 extractions. Table 2 presents summary statistics on household per-capita income as of June 2014 for all households in the registry as well as for all single adult households; Table 2 also presents statistics for beneficiary households who receive BF payments.

## III. Empirical Strategy

We now discuss our empirical strategy to estimate the bunching effect and jumping effect for the June 2014 BF reform. For simplicity, we focus on estimating the MVPF bounds in June 2016. Assuming that all behavioral responses to the June 2014 reform have occurred by June 2016 (which is reasonable given households are required to update their information every two years), the MVPF in June 2016 is a reasonable approximation for the long-term MVPF of this reform.[22] From the theoretical framework developed in Section I, in order to estimate MVPF bounds in June 2016, we need to form estimates of $B$ and $J$ in June 2016. More fundamentally, estimating these objects requires understanding the distribution of reported incomes under policy $\mathbf{p}$, $G(x; \mathbf{p})$, as well as the distribution of reported incomes under policy $\mathbf{p}'$, $G(x; \mathbf{p}')$ in June 2016. Hence, we now augment our model from Section I to include a simple notion of time.

Returning to our general model from Section I.C, let us suppose that the distribution of primitives, $F(\theta)$, is varying over time, which we denote by $F_t(\theta)$. We can also define the

[20] Moreover, the government aims to register all families with per-capita incomes below half the minimum wage (or total incomes below three times the minimum wage) (Veras Soares, 2011). The minimum wage was R\$724 per-month in 2014; thus half the minimum wage was R\$362, which is substantially higher than the BF threshold.

[21] Every time the MDS analyzes the Cadastro Único data, it creates one of these extractions. Therefore, the frequency of the extractions is a result of previous data analyses by the ministry. Appendix B.4 contains a figure depicting the timeline of the data extractions.

[22] We also calculate MVPF bounds for each month between June 2014 and June 2016 along with the cumulative MVPF for all months post-reform; the latter is computed using a discounted sum of jumping and bunching households over the entire time horizon a la Appendix A.8. Note the cumulative MVPF bounds are very similar to the June 2016 MVPF bounds given the assumption that all behavioral responses are observed by June 2016 (i.e., the number of bunching and jumping households post-June 2016 are equal to the numbers for June 2016). This is, however, an untestable assumption as there is another reform in June 2016 meaning we cannot estimate the impact of the June 2014 reform beyond June 2016. See Appendix C.1 for associated results and discussion.

Table 2— Summary Statistics as of June, 2014

| Variables | Mean | Median |
|---|---|---|
| Per Capita Income (single individual households) | 330.51<br>( 324.30) | 200.00 |
| Per Capita Income (all households) | 151.17<br>( 184.97) | 76.66 |
| Per Capita Income (single individual beneficiary households) | 39.73<br>( 39.72) | 40.00 |
| Per Capita Income (beneficiary households) | 56.20<br>( 53.24) | 47.00 |
| Observations (single individual households) | 4,232,528 | |
| Observations (all households) | 28,932,001 | |
| Observations (single individual beneficiary households) | 823,708 | |
| Observations (beneficiary households) | 13,124,018 | |

*Note:* This table shows summary statistics for households in the Cadastro Único database as of June, 2014. Per-capita income denotes household, monthly, per-capita income and is measured in Brazilian reais. The PPP conversion from US dollars to Brazilian reais was 1.813 in 2014 (OECD, `https://data.oecd.org/conversion/purchasing-power-parities-ppp.htm`).

number of households reporting an income under $z$ over time given a particular policy $\mathbf{p}$ as:

$$G_t(z;\mathbf{p}) = \int_{\theta:\hat{y}(\theta,\mathbf{p})\leq z} dF_t(\theta)$$

Let us denote June 2014 as $t = 0$. Returning to the specifics of the June 2014 BF reform, for single individual households, the reform changed the policy from $\mathbf{p} = \{b, \tau\} = \{70, 70\}$ to $\mathbf{p}' = \{b', \tau'\} = \{77, 77\}$. Plugging these values into Equations 8 and 9, our bounds on the MVPF in period $t$ are given by:

$$(11) \qquad MVPF_{L,t} \equiv 1 - \frac{77 \times [G_t(77;\mathbf{p}') - G_t(77;\mathbf{p})]}{77 \times G_t(77;\mathbf{p}') - 70 \times G_t(70;\mathbf{p})} = 1 - 77 \times \frac{J_t}{\text{Total Cost}_t}$$

$$(12) \qquad MVPF_{U,t} \equiv 1 + \frac{70 \times [G_t(70;\mathbf{p}) - G_t(70;\mathbf{p}')]}{77 \times G_t(77;\mathbf{p}') - 70 \times G_t(70;\mathbf{p})} = 1 + 70 \times \frac{B_t}{\text{Total Cost}_t}$$

Thus, to estimate the MVPF in a particular time period $t \geq 0$ (e.g., June 2016), we need to estimate the bunching effect and jumping effect at time period $t$, which, in turn, requires us to estimate the number of households locating below the old and new notch under both

$\mathbf{p}$ and $\mathbf{p}'$ at time $t$: $G_t(70;\mathbf{p})$, $G_t(70;\mathbf{p}')$, $G_t(77;\mathbf{p})$, and $G_t(77;\mathbf{p}')$.[23] Our fundamental identification challenge is that the BF reform was a national reform implemented uniformly across Brazil in June 2014: hence, we can observe $G_t(70;\mathbf{p}')$ and $G_t(77;\mathbf{p}')$ for $t \geq 0$ but we can only observe $G_t(70;\mathbf{p})$ and $G_t(77;\mathbf{p})$ for $t < 0$. For example, given that the distribution of household primitives may be changing over time, $G_t(77;\mathbf{p}') - G_{-1}(77;\mathbf{p})$ will conflate the true jumping effect, $G_t(77;\mathbf{p}') - G_t(77;\mathbf{p})$, with underlying time variation, $G_t(77;\mathbf{p}) - G_{-1}(77;\mathbf{p})$. Thus, we require a control group to estimate how the income distribution would have evolved from the period before the reform ($t = -1$ corresponding to May 2014) to time $t \geq 0$ *in absence of the reform*: $G_t(x;\mathbf{p}) - G_{-1}(x;\mathbf{p})$. In short, our identification strategy is going to rely on using regions of the reported income distribution that were not impacted by the reform as a control group to identify underlying time trends in the portions of the reported income distribution that were impacted by the reform. This brings us to our two identification assumptions.

### A. *Identification Assumption 1*

Our identification strategy relies on first finding a region of the reported income distribution which was not impacted by the reform. In the context of the baseline model in Section I.A, the number of single individual households reporting an income strictly below R\$70 should be unchanged by the reform. The intuition for this is that anyone who has a true income below R\$70 always reports truthfully both pre- and post-reform and anyone who has a true income above R\$70 prefers to misreport at the threshold rather than misreport to an income level below the threshold (because misreporting costs are increasing in the distance between true and reported incomes). While in this baseline model bunching should occur precisely at R\$70, in reality bunching is typically more diffuse due to optimization errors and/or frictions (Kleven, 2016). Our first identification assumption is that the number of people reporting incomes at or below R\$63 is unaffected by the reform. In essence, this assumption requires that the size of frictions is limited: for instance, we assume there are no households who would like to jump down to bunch at the new notch R\$77 but are forced to jump down to an income below R\$63 due to frictions (frictions are likely small given that we expect misreporting to be the predominant behavioral response in this high informality setting with eligibility based on self-reported income):[24]

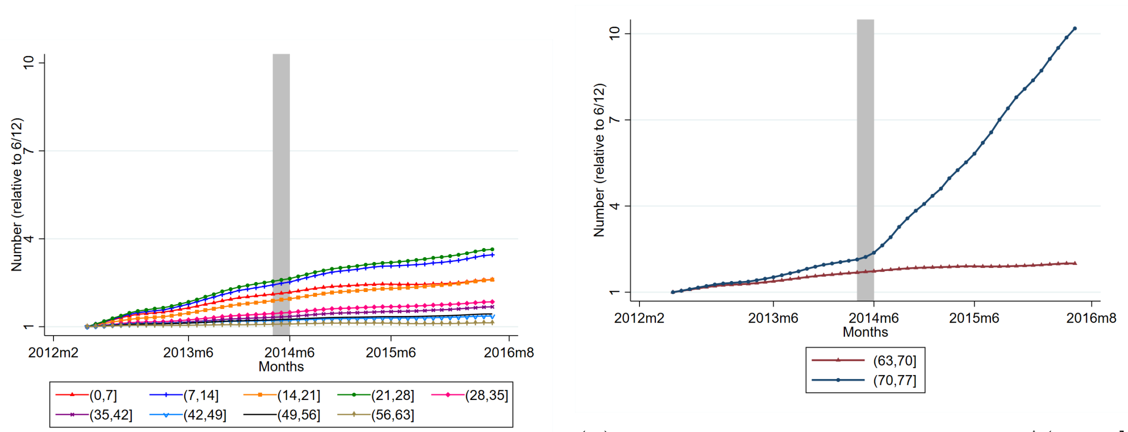
**Identification Assumption 1.** *The number of households reporting incomes below R\$63 is unaffected by the reform:* $G_t(x;\mathbf{p}) = G_t(x;\mathbf{p}')$ $\forall x \leq 63$, *t.*

To provide suggestive evidence that Assumption 1 is reasonable, Figure 3a plots the

[23] Recall that the total cost can be estimated from the components necessary to estimate the bunching effect and jumping effect as Total $\text{Cost}_t = 77G_t(77;\mathbf{p}') - 70G_t(70;\mathbf{p})$.

[24] Implicitly, Assumption 1 requires that households do not change their reported income in response to receiving a higher benefit, i.e., there are no income effects. While this is the case in the baseline misreporting model of Section I.A, it may not be the case in the more general model of Section I.C as households may reduce their labor supply and, in turn, their reported income in response to a higher benefit. Appendix C.2 argues that the lack of trend breaks in income bins under R\$63 (as in Figure 3a) along with a highly non-smooth initial reported income distribution suggests income effects are unlikely to be impacting the reported income distribution in our setting.

number of single individual households reporting in income bins of size 7 from R\$0 to R\$63 between June 2012 and June 2016 (with the numbers in each bin normalized to 1 in June 2012). Figure 3a shows that it is not obvious any of the bins below R\$63 were impacted by the reform.




(a) Number Reporting in Bins Below R\$63

(b) Number Reporting in R\$(63,70], R\$(70,77]

Figure 3. Number of Single Individual Households Reporting Incomes in Bins $\leq$ R\$77

*Note:* Panel (a) shows the number of single individual households with reported incomes in each seven increment bin below R\$63 between June 2012 and June 2016. Panel (b) shows the number of single individual households with reported incomes in R\$(63,70] and R\$(70,77]. The number in each bin is normalized to 1 in June 2012. For example, there were approximately 10 times as many single individual households with reported incomes in R\$$(70, 77]$ in June 2016 compared to June 2012. The timing of the reform (from the announcement in April 2014 to the enactment in June 2014) is indicated by the gray, shaded region.

Identification Assumption 1 implies that $J_t$ and $B_t$ can be expressed solely in terms of how the number of households reporting incomes in bins R\$$(63, 70]$ and R\$$(70, 77]$ changed as a result of the reform:

$$\begin{aligned} B_t &= -\left[G_t(70;\mathbf{p}') - G_t(70;\mathbf{p})\right] \\ &= -\left(\left[G_t(70;\mathbf{p}') - G_t(63;\mathbf{p}')\right] - \left[G_t(70;\mathbf{p}) - G_t(63;\mathbf{p})\right]\right) \end{aligned} \tag{13}$$

$$\begin{aligned} J_t =& G_t(77;\mathbf{p}') - G_t(77;\mathbf{p}) \\ =& [G_t(77;\mathbf{p}') - G_t(70;\mathbf{p}')] - [G_t(77;\mathbf{p}) - G_t(70;\mathbf{p})] + \\ & [G_t(70;\mathbf{p}') - G_t(63;\mathbf{p}')] - [G_t(70;\mathbf{p}) - G_t(63;\mathbf{p})] \end{aligned} \tag{14}$$

In particular, $B_t$ is equal to the reduction in households locating in R\$(63,70] at time $t$ as a result of the reform (i.e., the reduction in households bunching at and just below the old notch) while $J_t$ is equal to the increase in households locating in R\$(70,77] less the reduction in households locating in R\$(63,70] at time $t$ as a result of the reform (i.e., the increase in households bunching at and just below the new notch less the reduction in households bunching at and just below the old notch).

Figure 3b shows how the number of households reporting incomes in bins R$$(63, 70]$ and R$$(70, 77]$ evolved in a four year window around the reform from June 2012 to June 2016. In contrast to Figure 3a, Figure 3b shows a clear departure in trend for the bin R$$(70, 77]$ commensurate with reform. This provides highly suggestive evidence that the BF reform induced a substantial behavioral response that increased the number of households bunching at the new threshold. Note that the number of individuals reporting incomes in R$$(70, 77]$ is increasing over time prior to the reform - this is likely due to a combination of factors including population growth and (nominal) income dynamics.[25] In particular, Brazil was experiencing slowing and then negative per-capita GDP growth from 2012-2016 with a large drop in 2015 along with relatively high inflation (World Bank Data). However, the slope of the line for bin R$$(70, 77]$ in Figure 3b clearly changes at the time of the reform, leading to a sharp increase in the number of individuals locating in this bin. The fact that this reform induced a slope shift rather than a level shift may reflect households learning about the threshold shift over time. Alternatively, households may wait to update because they are only required to update every two years and/or to avoid suspicion of misreporting that could result from updating immediately after the reform.

The scale of Figure 3b makes it difficult to determine how the number of households in R$(63,70] changed as a result of the reform; Figure 13 in Appendix C.3 suggests that the number of households in R$(63,70] may have declined as a result of the reform although it is difficult to assess due to underlying time trends (Figure 14 in Appendix C.4 shows a much starker downward trend break for two adult households). Hence, we need a way to control for underlying time trends in the numbers reporting in R$$(63, 70]$ and R$$(70, 77]$ to estimate how these quantities would have evolved over time *in absence of the reform.* This brings us to our second identification assumption.

### B. *Identification Assumption 2*

At a high level, our second identification assumption is going to relate how the numbers reporting in bins below R$63 evolve over time to how the numbers reporting in our two bins of interest, R$$(63, 70]$ and R$$(70, 77]$, would have evolved over time in absence of the reform. In this sense, bins below R$63 can be viewed as our "control bins" while R$(70,77] and R$(63,70] can be viewed as our "treatment bins". Building towards our second identification assumption and our main empirical specification, consider the following regression, where $N_{(x-7,x],t}$ denotes the number of individuals reporting in income bin R$$(x - 7, x]$ in

[25]Similarly, the number of households on the Cadastro Único registry is growing over this time period - see Figure 16 in Appendix C.5. This too is likely due to a variety of factors including population growth, a struggling Brazilian economy, and increased awareness/understanding of the BF program over time.

month $t$:

$$\underbrace{\log\left(N_{(x-7,x],t}\right)}_{\log \# \text{ in } (x-7,x]} = \underbrace{\sum_{k=0}^{K} \zeta_{k,x} t^k}_{\text{bin-specific polynomial}} + \underbrace{\kappa_{1,x} post_t + \kappa_{2,x} post_t \times t}_{\text{post-reform deviation from polynomial}} + \nu_{x,t} \tag{15}$$

where $post_t$ takes value 1 if month $t$ is after the reform and 0 otherwise (i.e., $post_t = 1$ if $t \geq$June 2014 and 0 otherwise). Regression 15 simply estimates a break from the (bin-specific) polynomial time trend in the post-reform period for the (log) number of people reporting in R\$$(x-7,x]$. Running Regression 15 separately for each bin (using cubic bin-specific polynomials), Figure 4 plots $\hat{\kappa}_{1,x} + \hat{\kappa}_{2,x}\bar{t}$ for each $x \in \{7, 14, ..., 77\}$ where $\bar{t}$ represents the final month in our analysis period (June 2016). In other words, Figure 4 plots how the log number of people reporting in each seven increment bin below R\$77 deviated from its bin-specific cubic time trend in the post-reform period:

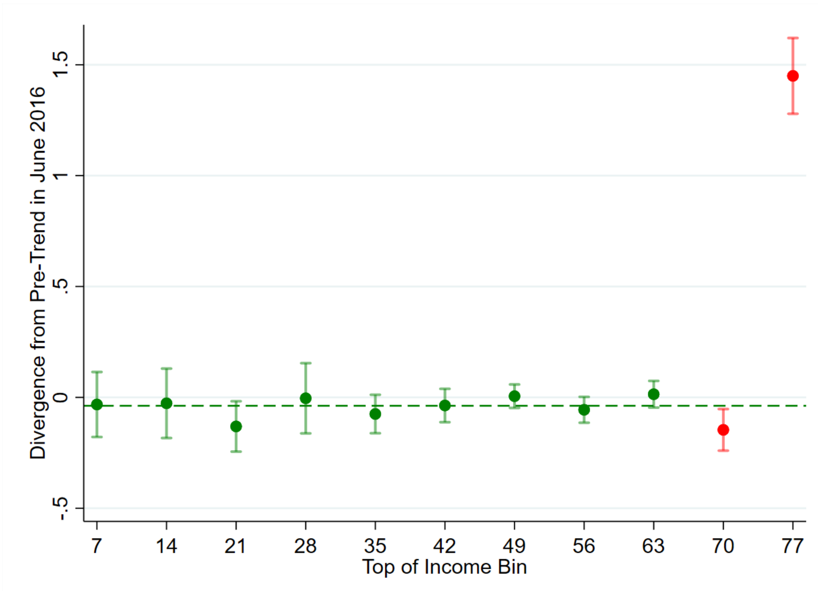



Figure 4. Deviation from Cubic Time Trend in Post-Reform Period by Income Bin

*Note:* This figure plots how the log number of people reporting in each seven increment bin deviated in June 2016 from a bin-specific, cubic time trend estimated via Regression 15 using monthly reported income data from June 2012 to June 2016 along with 95% confidence intervals using robust standard errors. For each $x \in \{7, 14, ..., 77\}$, we estimate the deviation from the bin-specific time trend in June, 2016 as $\hat{\kappa}_{1,x} + \hat{\kappa}_{2,x}\bar{t}$, where $\bar{t}$ represents the final month in our analysis period (June 2016). The green horizontal line plots the average deviation (-0.038) from trend for bins with $x \leq R\$63$.

Figure 4 shows that the number of people reporting in income bins below R\$63 saw very minor (and mostly statistically insignificant) trend breaks in the post-reform period. The dashed green line in Figure 4 indicates an average trend break of -0.038 across these bins, indicating that the number of people reporting incomes in bins $\{R\$(0, 7], R\$(7, 14], ..., R\$(56, 63]\}$ was, on average, approximately 3.8% lower in June, 2016 relative to polynomial trend. However, under Identification Assumption 1, any post-reform trend break for these bins must be due to underlying time variation unrelated to the reform. Loosely speaking, our second identification assumption is that our two treatment bins, R$(63, 70] and R$(70, 77], would have seen the same deviation post-reform as the control bins had the reform not happened, i.e., these two bins also would have seen a 3.8% reduction in the number of people in June, 2016 (relative to their bin-specific polynomial time trends) had the reform not happened. However, as can be seen in Figure 4, the number of people reporting incomes in R$(63, 70] saw a reduction of around 13.6% (a 0.146 log-point decrease) while the number of people reporting incomes in R$(70, 77] saw an increase of 326.3% (a 1.45 log-point increase) as of June 2016. Formally, our second identification assumption is:

**Identification Assumption 2.** *In absence of the reform, the (log) number of people reporting in each bin evolves according to:*

$$\log\left(N_{(x-7,x],t}\right) = h(t) + \sum_{k=0}^{K} \alpha_{k,x} t^k + \epsilon_{x,t} \quad \textit{for } x \in \{7, 14, ..., 77\}$$

*where* $\mathbb{E}[\epsilon_{x,t}|t] = 0$.

Under Identification Assumption 2, the difference between the (log) number of households in any two 7-increment bins below R$77 is, *in absence of the reform*, governed by a stable polynomial plus a random error term. Combining Assumptions 1 and 2, we can use our control bins (i.e., bins ≤R$63) to identify $h(t)$ in the post-reform period.[26] In other words, we can use our control bins to identify the expected deviation from bin-specific polynomials in the post-reform period if the reform did not occur. Any deviation observed above and beyond $h(t)$ in our two treatment bins is then attributed to the reform. This brings us to our main empirical specification: a generalized difference-in-difference specification where we allow for flexible pre-treatment dynamics between treatment and control bins:

$$\log\left(N_{(x-7,x],t}\right) = \delta_t + \sum_{k=0}^{K} \alpha_{k,x} t^k + [\beta_{1,x} post_t + \beta_{2,x} post_t \times t] \times treat_x + \epsilon_{x,t} \text{ for } x \in \{7, 14, ..., 77\} \tag{16}$$

where $\delta_t$ represents a set of month fixed-effects; $treat_x$ takes value 1 if $x \in \{70, 77\}$ and 0 otherwise; and $\sum_{k=0}^{K} \alpha_{k,x} t^k$ captures polynomial time trends for each bin (indexed by

[26] We use the distribution below R$63 to control for underlying time trends consistent with Kleven (2016), who suggests using data below the notch to estimate bunching when extensive margin responses are present. In theory, we could also use the distribution *sufficiently far* above R$77 to control for underlying time trends as this region is also unimpacted by the reform. Kleven and Waseem (2013) discuss how to identify an unimpacted region of the density sufficiently far above a notch by equating the excess mass below the notch with the missing mass above the notch. However, our setting likely features extensive margin responses: some of the "excess" households locating below R$77 post-reform may have not reported an income (and thus not be on the registry) in absence of the reform. Thus, we cannot equate the excess mass below R$77 with the missing mass above R$77.

$x$) which predate the reform and are assumed to persist into the post-reform period had the reform not occurred. Under our two identification assumptions, the causal impact of the reform on the (log) number of households locating in R$\$(x-7,x]$ in month $t$, denoted $\Delta\log\left(N_{(x-7,x],t}\right)$, is equal to $\hat{\beta}_{1,x}+\hat{\beta}_{2,x}t$ for $x\in\{70,77\}$.

Before we present the results from Equation 16 in Section IV (using various polynomial degrees $K$), let us discuss the statistical and economic interpretations of Identification Assumption 2. From a statistical perspective, Identification Assumption 2 is simply a generalization of the standard difference-in-difference assumption (this generalization is employed in, for example, Wolfers (2006) and is discussed more generally in Mora and Reggio (2013)). Setting $K=0$ results in a standard difference-in-difference estimator in which the control groups and treatment groups are assumed to have "parallel trends" pre-reform that would have persisted in the post-reform period had the reform not occurred. Larger values of $K$ require progressively weaker parallel assumptions. For instance, setting $K=1$ requires the "parallel growths" assumption which asserts that the growth rates in the treatment and control groups over time are the same (i.e., that second and higher differences between treatment and control groups are constant over time).[27] Setting $K>1$ results in what Mora and Reggio (2013) refer to as the "parallel-$K$" assumption which asserts that the $K+1$ and higher differences between the treatment and control groups are constant over time.

But what is the economic meaning of Identification Assumption 2? Recall that in light of Identification Assumption 1, our fundamental identification challenge is that we do not know how the number of households locating in the two treatment bins, R$\$(63,70]$ and R$\$(70,77]$, would have evolved over time in absence of the reform due to changes in the underlying distribution of household primitives (e.g., endowed incomes, labor productivities, preferences over misreporting etc.) By making Identification Assumption 2, we assume that the underyling distribution of primitives is evolving such that the differences over time between the number of people reporting in each income bin are well approximated by polynomials which would have persisted in absence of the reform. Note, every difference-in-difference study implicitly requires an assumption like Identification Assumption 2 that ensures the relationships between treatment and control groups are stable over time (in absence of reforms). Identification Assumption 2 is not fully testable in the same way that the standard differences-in-differences assumption (that parallel pre-trends persist post treatment) is not fully testable. However, as in the case of the standard "parallel trends" assumption, we can gauge whether Identification Assumption 2 seems sensible by looking at pre-reform data. Appendix C.6 shows how the differences $\log\left(N_{(70,77]}\right)-\log\left(N_{(x-7,x]}\right)$ and $\log\left(N_{(63,70]}\right)-\log\left(N_{(x-7,x]}\right)$ evolved in the pre-reform period for $x\in\{7,14,...,63\}$. These differences follow stable, low order polynomial relationships in the pre-period. While this of course does not imply that these relationships would have persisted into the post-reform period (just as parallel pre-trends do not imply that the trends would continue on a parallel path post treatment), it is at least suggestive that stable relationships exists

[27] A number of papers discuss the extended version of differences-in-differences under the "parallel growths" assumption (e.g., Mora and Reggio (2013), Bilinski and Hatfield (2019), and Rambachan and Roth (2020)).

between the various income bins. Moreover, because we have several control bins, we can *partially* test the validity of our two identification assumptions. In particular, we can run placebo tests where we pretend that some of our control bins are impacted by the reform and use the remaining control bins to estimate these placebo "treatment effects". Results from these placebo tests are presented at the end of Section IV.

### C. Could We Use a Bunching Estimator to Estimate $B$ and $J$?

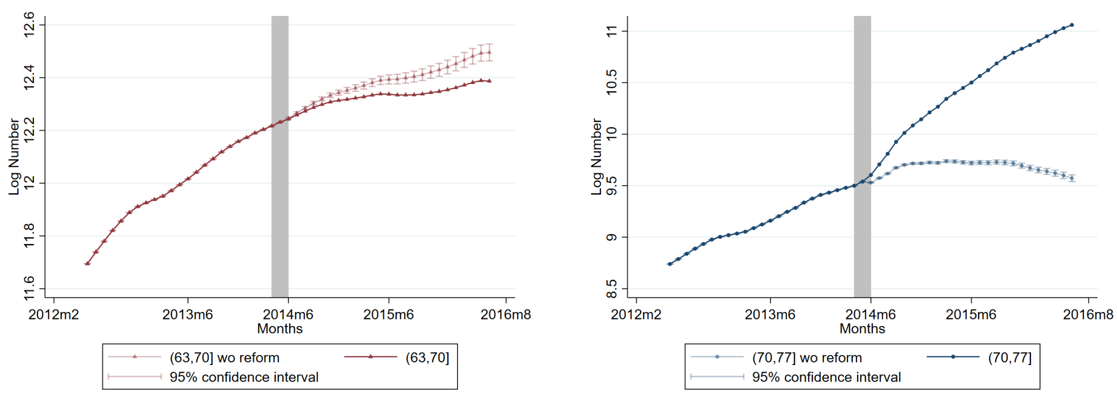
We use a difference-in-difference strategy to estimate the reduction in bunching at the old notch and the increase in bunching at the new notch as a result of the reform. But one may wonder whether we could have used standard bunching techniques (developed in Kleven and Waseem (2013) and Saez (2010)) to back out the changes in these masses. Of course, even using standard bunching techniques, one must account for time variation by estimating how bunching would have changed in absence of the reform. For instance, one could make standard smoothness assumptions to estimate a counterfactual income distribution that would ensue in absence of the notch both pre- and post-reform. By assuming that the relationship between the counterfactual and observed reported income distributions in the pre-reform period would have stayed the same in the post-reform period in absence of the reform, one could estimate how bunching would have changed over time due to time trends; taking the observed difference in bunching minus the difference in bunching due to time trends would yield an estimate of the change in bunching due to the reform. A similar strategy is employed in Carril (2022). However, these sorts of standard bunching methods are unlikely to be feasible in our setting. As seen in Figure 20 in Appendix C.7, the reported income distribution in Brazil is *extremely* non-smooth. For instance, in June 2014, in addition to substantial bunching at the notch of R$70, there is extreme bunching at numbers equal to 0 mod 50, less severe bunching at numbers equal to 0 mod 10, and substantial bunching at R$60 (R$60 was a previous BF eligibility threshold implemented in 2006 while the notch of R$70 was implemented in 2009). While bunching estimators have been augmented to deal with "round-number" bunching and other "reference-point" bunching (e.g., Kleven and Waseem (2013) and Best and Kleven (2017)), we believe that the pervasiveness and the variability of round-number and reference-point bunching in our setting will make it too difficult to precisely identify the counterfactual distributions around our notches. Thus, our empirical strategy highlights an alternative way to estimate changes in bunching when smoothness assumptions cannot be used to estimate counterfactual distributions.

## IV. Results

In this section, we present results from our generalized difference-in-difference specification, Equation 16, and, in turn, calculate the bunching effect and jumping effect along with our bounds on the MVPF. Based on these bounds, we then discuss the welfare implications of the reform. We then discuss robustness of our results.

### A. *Difference-in-Difference Results*

First, we present results from estimating Equation 16 setting $K = 3$, which assumes the difference in the number of households reporting in any two income bins below R\$77 approximately follows a cubic polynomial over time (in absence of the reform). Once we have estimated Equation 16, we can recover the causal impact of the reform on the log number of households in R$(x-7, x]$ for $x \in \{70, 77\}$ in any given month $t$, denoted $\Delta \log\left(N_{(x-7,x],t}\right) = \hat{\beta}_{1,x} + \hat{\beta}_{2,x}t$. Figure 5 plots the evolution of $\log\left(N_{(63,70],t}\right)$ and $\log\left(N_{(70,77],t}\right)$ over time along with the estimated counterfactual path if the reform did not occur: $\log\left(N_{(63,70],t}\right) - \Delta \log\left(N_{(63,70],t}\right)$ and $\log\left(N_{(70,77],t}\right) - \Delta \log\left(N_{(70,77],t}\right)$. The difference between the actual path and the counterfactual path yields the causal impact of the reform under our two identification assumptions.




(a) $N_{(63,70]}$: Actual & Counterfactual (b) $N_{(70,77]}$: Actual & Counterfactual

Figure 5. Causal Impact of the Reform Estimated From Equation 16

*Note:* This figure shows the log number of single individual households reporting incomes in R$(63, 70]$ and R$(70, 77]$ from June 2012 to June 2016 along with the counterfactual paths had the reform not happened. The counterfactual paths are equal to the actual log number of people reporting in the given interval minus the causal impact of the reform, $\hat{\beta}_{1,x} + \hat{\beta}_{2,x}t$, estimated using Equation 16 where we set $treat_x = 1$ if $x \in \{70, 77\}$ and $K = 3$. Confidence intervals are constructed from clustered standard errors at the bin level. The timing of the reform is indicated by the gray, shaded region.

Consistent with the theoretical model from Section I.A, Figure 5 shows that the reform led to a large, significant increase of about 1.5 log points in the number of households reporting in bin R$(70, 77]$ and a significant decrease of about 0.11 log points in the number of households reporting in bin R$(63, 70]$.[28] Translating these log changes into levels, the reform led to an increase of approximately 49,000 households locating in R$(70, 77]$ and

[28] All standard errors for our analysis are clustered at the bin level (using STATA's default small-number-of-clusters bias adjustment) as this is the level of "treatment assignment", see Abadie et al. (2017). Thus, we have 9 control

a decrease of approximately 27,000 households locating in R$(63, 70].[29] Plugging these numbers into Equations 13 and 14 and denoting June 2016 as $\bar{t}$, we get that $B_{\bar{t}} \approx 27,000$ and $J_{\bar{t}} \approx 22,000$. Thus, of the additional 49,000 households bunching at the new notch, 27,000 would have bunched at the old notch had the reform not happened, while 22,000 would have located above the new notch had the reform not happened (i.e., 22,000 households jump into the program as a result of the reform).[30]

We repeat this exercise under the alternative assumptions that the differences in the (log) numbers in any two bins follow quadratic, quartic, or quintic polynomials over time, i.e., we re-estimate Equation 16 for $K = 2, 4, 5$.[31] Figures of the counterfactual paths for our two treatment bins under these alternative assumptions are shown in Appendix C.10. Table 3 presents the estimated impact of the reform on the numbers locating in R$(63,70] and R$(70,77], $\Delta N_{(63,70],\bar{t}}$ and $\Delta N_{(70,77],\bar{t}}$, along with the estimated bunching effect $B_{\bar{t}}$, estimated jumping effect $J_{\bar{t}}$, and lower and upper bounds on the MVPF in June 2016. All of these quantities are stable across different values of $K$.

### *B. Discussion*

Columns (5) and (6) of Table 3 show that our lower bound on the MVPF of the June 2014 BF reform is approximately 0.90 whereas our upper bound on the MVPF is approximately 1.12. Notably, these bounds are somewhat tightly centered around 1. This is because the number of mechanical households (households locating below R$70 both with and without the reform) is large relative to $B$ and $J$: in June 2016, we estimate there to be around 1.88 million mechanical households.

What can we say about the welfare implications of an MVPF bounded between 0.90 and 1.12? We can be sure that the reform was welfare improving as long as the government values giving R$0.90 to BF households in a non-distortionary manner more than spending R$1 on their next best alternative. On the other hand, the reform was welfare decreasing

clusters and 2 treatment clusters. However, Imbens and Kolesár (2016) show that these standard errors are modestly under-estimated when the number of clusters is small (they find that for samples with 10 clusters, 95% confidence intervals only have a 91% coverage rate). Hence, our standard errors may be modestly overstating the true statistical confidence we have in our estimates. An alternative approach is to use non-clustered wild bootstrapped p-values (note, wild *cluster* bootstrapped p-values lead to severe bias in difference-in-difference settings with a small number of treated clusters, see Roodman et al. (2019)). In our setting, these p-values are smaller than those generated from our cluster-robust standard errors. Finally, Ferman and Pinto (2019) suggest another alternative for difference-in-difference settings with a small number of treated clusters; unfortunately, their results rely on asymptotic theory in the number of control clusters, which we believe is inappropriate for our setting given that we only have 9 control clusters.

[29] In Figure 5b, we estimate that the number of households in R$(70, 77] would have slightly declined over the first six months of 2016 had the reform not occurred (this results from the relative concavity between the R$(70, 77] bin and our control bins in the pre-reform period). This does not significantly impact our results: we show in Appendix C.8 that our main results are very similar even if we make the conservative assumption that R$(70, 77] would have (counterfactually) grown at the maximum rate observed for any control bin over the post-reform period.

[30] Technically, some of these 22,000 jumpinghouseholds may have responded on the extensive margin, i.e., in absence of the reform, they may have not have reported an income to the registry. We provide descriptive evidence to suggest that a number of these jumping households are likely responding on the extensive margin - see Appendix C.9.

[31] We do not estimate Equation 16 for $K = 0$ or $K = 1$ as pre-trends between treatment and control bins are often not constant or linear (see Figures 17 and 18 in Appendix C.6).

Table 3— Impacts of Reform and MVPF Bounds in June 2016 Estimated From Equation 16

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Polynomial Degree, $K$ | $\Delta N_{(63,70],\bar{t}}$ | $\Delta N_{(70,77],\bar{t}}$ | $B_{\bar{t}}$ | $J_{\bar{t}}$ | $MVPF_{L,\bar{t}}$ | $MVPF_{U,\bar{t}}$ |
| Quadratic, $K=2$ | -26,279 | 51,759 | 26,279 | 25,480 | 0.88 | 1.11 |
| | (6,163) | (2,160) | (6,163) | (6,659) | (0.03) | (0.03) |
| Cubic, $K=3$ | -27,452 | 49,247 | 27,452 | 21,794 | 0.90 | 1.12 |
| | ( 4,357) | (234) | (4,357) | (4,592) | (0.02) | (0.02) |
| Quartic, $K=4$ | -29,338 | 50,873 | 29,338 | 21,535 | 0.90 | 1.13 |
| | ( 6,257) | (1,345) | (6,257) | (6,503) | (0.03) | (0.03) |
| Quintic, $K=5$ | -29,240 | 50,559 | 29,240 | 21,318 | 0.90 | 1.13 |
| | ( 5,912) | ( 1,184) | ( 5,912) | ( 6,141) | ( 0.03) | ( 0.03) |

*Note:* Columns (1) and (2) show the estimated impacts of the reform on the number of single individual households reporting incomes in bins R$(63,70] and R$(70,77] for June 2016: $\Delta N_{(63,70],\bar{t}}$ and $\Delta N_{(70,77],\bar{t}}$. Estimates are calculated using Equation 16 with various polynomial degrees $K \in \{2,3,4,5\}$. Columns (3) and (4) show the estimated bunching effect and jumping effect for June 2016, $B_{\bar{t}}$ and $J_{\bar{t}}$, calculated using Equations 13 and 14. Columns (5) and (6) show the estimated upper and lower bounds for the MVPF for June 2016, calculated using Equations 11 and 12. Standard errors are presented in parentheses and are computed via the delta method from the clustered standard errors estimated in Equation 16.

if the government values spending R$1 on their next best alternative more than giving R$1.12 to BF households in a non-distortionary manner.

But what is the government's best alternative use of funds? In general, this is a difficult question to answer. Lindert et al. (2007) show that BF performs better (in terms of targeting performance) than other anti-poverty programs in both Brazil and other countries in Latin America; this suggests that there is not a better alternative anti-poverty program that the government could finance. Instead, for expositional simplicity we assume that the next best use of funds is a non-distortionary universal basic income (UBI). Under this assumption, we can perform a conservative back-of-the-envelope calculation to argue that the 2014 BF reform was almost certainly welfare improving. Suppose the government is utilitarian and that the true income distribution is log-normal with a mean per-capita income of R$615/month and a standard deviation of R$844/month (taken from the World Bank's PovcalNet database for Brazil in 2016). Conservatively, we assume that the BF household per-capita income distribution matches the distribution for the bottom 50% of income earners in Brazil.[32] Moreover, we assume utility of consumption is CRRA: $u(c) = c^{1-\gamma}/(1-\gamma)$. Based on these assumptions, the government prefers giving R$0.90

[32] Bastagli (2008) finds that of all transfers paid in 2004, 91% went to households in the bottom 50% of the income distribution. Lindert et al. (2007) suggests even better targeting performance, finding that 94% of all benefits go to the bottom 40%. Notably, transfers are not evenly distributed across the bottom 50% of households: the poorest households within this group receive more (e.g., Lindert et al. (2007) find that the bottom 20% of the income distribution receive 73% of BF transfers). Thus, we believe the assumption that the income distribution of BF households is the same as the income distribution for the bottom 50% of income earners in Brazil is highly conservative.

(in a non-distortionary manner) to BF households more than spending R$1 on a non-distortionary UBI as long as $\gamma > 0.14$. While estimates for $\gamma$ vary widely, the vast majority of estimates are bigger than 0.14 (Chetty (2006), Outreville (2014), Bergstrom and Dodds (2023)). Alternatively, if $\gamma = 1$ (i.e., $u(c) = \log(c)$), giving R$0.90 (in a non-distortionary manner) to BF households yields equivalent welfare gains to spending R$1.50 on a non-distortionary UBI. Thus, if the next best alternative policy is a non-distortionary UBI and the government is utilitarian, the BF reform was almost certainly welfare improving. This finding contributes to the debate around targeted vs. universal transfers in developing settings (Hanna and Olken, 2018): we find that even in a setting with a highly pronounced notch and substantial scope for misreporting, the efficiency cost generated by behavioral responses is simply not large enough to outweigh the equity gain associated with increased benefit generosity targeted to poor households (relative to a universal transfer).

Finally, when calculating our bounds for the MVPF, we have abstracted from a couple of complexities in the policy environment. First, we have ignored the impact that behavioral responses to the BF reform had on other components of the government's budget. There is evidence to suggest that some beneficiaries partially substitute formal sector employment for informal sector employment to become eligible for the BF benefit (as informal income is less easily verified making misreporting less costly; see De Brauw et al. (2015)). While it is easier to evade taxes in the informal sector, it is unlikely that this behavioral response will affect income tax revenues as BF beneficiaries are almost certainly sufficiently poor so as to be exempt from income taxation.[33] However, such behavioral responses may lead to reductions in payroll tax revenues (as argued by Bergolo and Cruces (2021) in the context of a cash transfer program in Uruguay). A reduction in payroll tax revenue would increase the total cost of the reform leading to a reduction in both our lower and upper bounds for the MVPF (see footnote 16). However, there is an offsetting effect: Gerard, Naritomi and Silva (2021) recently showed that a reform to BF in 2009, which led to an increase in transfers, also led to a rise in formal sector employment of non-beneficiaries. They show that this offsetting effect dominates, generating a net positive impact on overall formal sector employment and tax revenue. Thus, we conjecture that the June 2014 BF reform may have had a small, positive impact on formal sector employment, generating a modest positive fiscal externality.[34] A positive fiscal externality would raise our estimates for both the lower and upper bound of the MVPF, and would, thus, only reinforce our finding that the 2014 reform was welfare improving relative to a non-distortionary UBI.

Second, as mentioned in Section II, not all eligible households receive the BF benefit (due to municipality-level quotas on the number of beneficiaries). We discussed an extension in Section I.C that showed that our MVPF bounds are robust to eligible households only receiving the benefit with some probability. However, this extension assumed that

[33] For example, in 2015, individuals earning less than R$1903.98 per-month were exempt from income taxation.

[34] As discussed in Appendix B.3, most federal social programs in Brazil have substantially higher income thresholds than BF. Hence, the June 2014 BF reform would not have induced fiscal externalities on other social programs as long as the households who changed their reported income in response to the reform did not have inordinately large jumps (i.e., in absence of the reform, these households would have reported below the thresholds of these other programs).

the probability of receiving the benefit does not vary across households (conditional on reporting an income below the threshold); this allows us to cancel out the probability in both the numerator and denominator of the MVPF. Fortunately, this appears to be the case for the BF reform: using data from the Cadastro Único registries on which eligible households are BF beneficiaries, we see that in June 2016, 77.6% of those reporting at or below R$77 are beneficiaries with this percentage mostly constant across all eligible bins, suggesting that the probability of receiving the benefit is roughly constant across groups of households (see Figure 28 in Appendix C.11). This percentage is very similar to Gerard, Naritomi and Silva (2021) who find that in 2010, 79% of households reporting below the extreme-poverty threshold are BF beneficiaries.

### C. *Robustness to Identification Assumption 1*

We now explore robustness of our main results. First, Identification Assumption 1 may not hold if optimization frictions are large. For example, consider households who, in response to the reform, jump below the threshold by changing their labor supply but face large labor market frictions. These households may not be able to perfectly jump and bunch at the new notch, but may instead jump to an income level below R$63. Hence, large labor market frictions may imply that the distribution below R$63 is impacted by the reform. To test for this possibility (or equivalently, to allow for larger frictions), we relax Identification Assumption 1 by instead assuming that the number of people reporting in bins ≤R$56 are unaffected by the reform, allowing for the possibility that the number reporting in bin R$$(56, 63]$ is affected by the reform. We augment Equation 16 by setting $treat_x = 1$ if $x \in \{63, 70, 77\}$ and 0 otherwise, i.e., we include R$$(56, 63]$ as a treatment bin. Figure 6 plots the actual and counterfactual paths for $\log\left(N_{(56,63],t}\right), \log\left(N_{(63,70],t}\right)$, and $\log\left(N_{(70,77],t}\right)$. It does not appear that the reform had a significant impact on the number reporting in R$$(56, 63]$ and the impact of the reform on our two original treatment bins is robust to including R$(56,63] as a treatment bin. Table 8 in Appendix C.12 presents results for this alternative specification.

### D. *Robustness to Identification Assumption 2*

Identification Assumption 2 essentially boils down to assuming that the differences over time between the (log) number of people reporting in each income bin are well approximated by polynomials which would have persisted in absence of the reform. Identification Assumption 2 therefore implies that if, for example, bin R$(35,42] experiences a 1% deviation from its bin-specific polynomial trend, bin R$(63,70] would also experience a 1% deviation from its bin-specific polynomial trend in absence of the reform. However, one may believe that, for example, a structural change in the economy which leads to a 1% deviation from the bin-specific polynomial time trend for bin R$(35,42] would lead to a 2% deviation for bin R$(63,70]. As such, we relax Identification Assumption 2 to instead

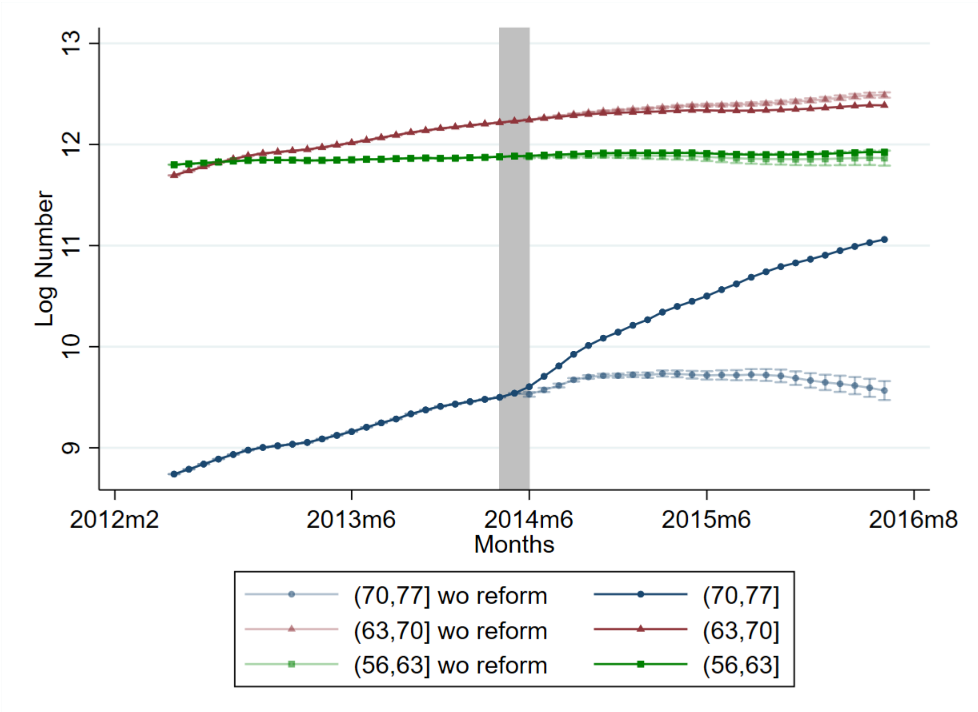



Figure 6. Counterfactual Paths for Three Treatment Bins Using Equation 16

*Note:* This figure shows the log number of single individual households reporting incomes in R$(56, 63], R$(63, 70], and R$(70, 77] from June 2012 to June 2016 along with the counterfactual paths had the reform not happened. The counterfactual paths are estimated using Equation 16 (where we set $treat_x = 1$ if $x \in \{63, 70, 77\}$, i.e., we have three treatment bins: R$(56, 63], R$(63, 70], and R$(70, 77]). We assume the difference in the (log) number reporting in any two bins follows a cubic time trend in absence of the reform (i.e., we set $K = 3$ in Equation 16). Confidence intervals are constructed from clustered standard errors at the bin level. The timing of the reform is indicated by the gray, shaded region.

assume that the log number reporting in each bin evolves according to:

$$\log\left(N_{(x-7,x],t}\right) = \gamma_x h(t) + \sum_{k=0}^{K} \alpha_{k,x} t^k + \nu_{x,t} \quad \text{for } x \in \{7, 14, ..., 77\}$$

Thus, we now assume that deviations from underlying bin-specific polynomials (in logs or, equivalently, percentage terms) are not necessarily the same across all bins. This relaxation of Identification Assumption 2 allows for a structural change in the economy which creates a 1% deviation from bin-specific polynomial trend for R$(35,42] to create a $(\gamma_{70}/\gamma_{42})$% deviation from bin-specific polynomial trend for bin R$(63,70]. Under this relaxed identification assumption (which nests Identification Assumption 2), we estimate the following non-linear least squares regression:

$$\log\left(N_{(x-7,x],t}\right) = \gamma_x \delta_t + \sum_{k=0}^{K} \alpha_{k,x} t^k + \left[\beta_{1,x} post_t + \beta_{2,x} post_t \times t\right] \times treat_x + \epsilon_{x,t} \text{ for } x \in \{7, ..., 77\} \tag{17}$$

Essentially, running Regression 17 will estimate a common set of month dummies, $\delta_t$, as well as bin-specific factors, $\gamma_x$, which multiply these common month dummies for each bin $x$. We show in Appendix C.13 that our results are robust to this relaxation of Identification Assumption 2.

### E. *Robustness to Household Composition*

Our main analysis is restricted to single individual households due to the fact that the schedule for households with more than one individual features both a notch at R$70 as well as a kink below R$70 (e.g., this kink is at R$35 for two adult households with no children; see Section II). Consequently, the 2014 BF reform led to both a change in the notch *and* a change in the kink for households with more than one individual. Thus, for these households, it is not necessarily reasonable to make Identification Assumption 1 as we might expect the reform to affect the number of people reporting in bins around the kink. However, many studies have shown that behavioral responses to kinks are typically small; kinks generally induce substantially less bunching than do notches (Kleven, 2016). Hence, we estimate MVPF bounds for two adult households without children by just ignoring the presence of the kink and using all income bins below R$63 as control groups just as in our main analysis, noting that our identification assumptions are imperfect for these households. For two adult households without children we see the same pattern of behavioral responses to the reform and estimate lower bounds for the MVPF between 0.88 and 0.96 and upper bounds for the MVPF between 1.08 and 1.12; see Appendix C.4.[35]

### F. *Placebo Tests*

We can *partially* test the validity of our two identification assumptions via placebo tests. Together, Identification Assumptions 1 and 2 imply that all of the income bins below R$63 should evolve approximately according to a common time trend plus a bin-specific polynomial throughout the entire analysis period. To test this, we re-estimate Equation 16 but only include bins below R$63 (i.e., $x \in \{7, 14, ..., 63\}$) and randomly assign some of these bins to be "treatment" bins (i.e., $treat_x = 1$). If both of our identification assumptions

[35] In Appendix C.14, we show strong suggestive evidence of a behavioral response to the change in the basic benefit notch for households with children, but we do not attempt to bound the MVPF for these households. This is because households with children may also receive the "variable benefit". Both the level and location of the notch associated with the variable benefit also changed with the June 2014 reform. Thus, the WTP of households with children needs to account for changes in the variable benefit schedule in addition to changes in the basic benefit schedule. This exercise is beyond the scope of the current paper.

are valid, the "treatment effects" from these placebo regressions should be close to zero, i.e., the post-reform deviation of the "treated" bins relative to the post-reform deviation of the "control" bins should be close to 0. There are a total of $\sum_{i=1}^{8}\binom{9}{i} = 510$ different ways to assign our nine bins below R\$63 "treatment" status.

Figure 7 plots the results from all 510 such regressions, showing the estimated "treatment effects" in June 2016: $\hat{\beta}_{1,x} + \hat{\beta}_{2,x}\bar{t}$ for $x \in \{7, .., 63\}$, where $\bar{t}$ represents June 2016. Each bin gets assigned $treat_x = 1$ a total of 255 times meaning we have 255 "treatment effect" estimates for each bin. Figure 7 shows a fairly tight clustering of these treatment effects around zero: the mean "treatment effect" (in absolute value) across all bins is around 4% (0.04 log points). Hence, the "treatment effects" are relatively small for bins below R\$63.[36] This suggests that Identification Assumptions 1 and 2 are reasonable.

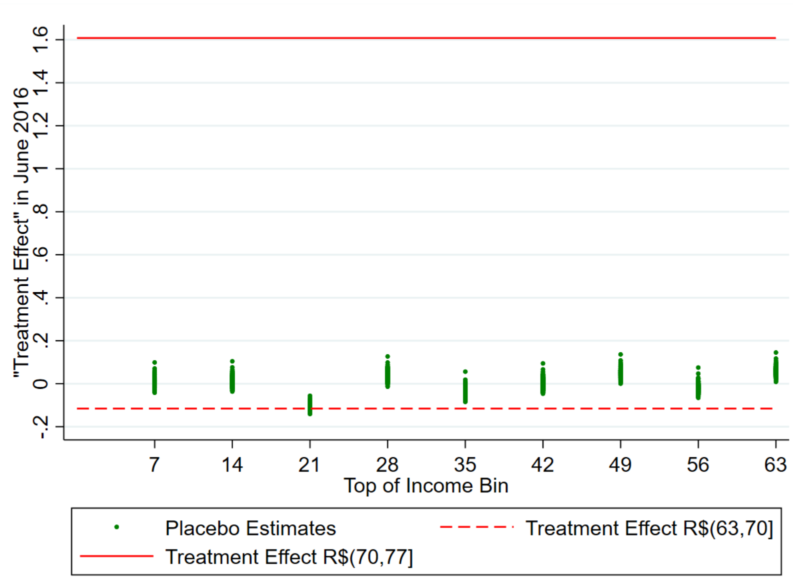



Figure 7. Placebo Analysis: Calculating "Treatment Effects" for each Control Bin

*Note:* This figure shows the placebo "treatment effects" (in logs) for each income bin R\$$(x-7, x]$ obtained from estimating Regression 16 with only bins $x \in \{7, 14, ..., 63\}$ and every possible assignment of $treat_x \in \{0, 1\}$ for each $x \in \{7, 14, ..., 63\}$. There are $510 = \sum_{i=1}^{8}\binom{9}{i}$ such regressions and we plot the values of $\hat{\beta}_{1,x} + \hat{\beta}_{2,x}\bar{t}$ for each bin R\$$(x-7, x]$ whenever bin R\$$(x-7, x]$ is assigned to be a treatment bin ($\bar{t}$ represents June, 2016). We use bin-specific cubic polynomials in each regression, i.e., $K = 3$. The solid red line shows the treatment effect for R\$(70,77] and the dashed red line shows the treatment effect for R\$(63,70] when $K = 3$.

Finally, we can also use these placebo tests as an alternative gauge of the statistical

[36] We find a negative "average treatment effect" of -0.105 log points for R\$$(14, 21]$. However, R\$$(14, 21]$ seems implausibly far away from the notch to be impacted by the reform, especially given that other bins around R\$(14,21] do not appear to be impacted by the reform. Hence, we interpret this negative "average treatment effect" simply as random variation.

significance of our estimated treatment effects for income bins R$(70,77] and R$(63,70] from Regression 16. Under Identification Assumptions 1 and 2, the post-reform deviations from bin-specific trend plotted in Figure 7 are due to randomness. Hence, the distribution of "treatment effects" in Figure 7 can be used to construct p-values for the estimated treatment effects for income bins R$(70,77] and R$(63,70] from Regression 16 in a manner akin to randomization inference.[37] In Figure 7, the solid red line shows the treatment effect of about 1.5 log points for R$(70,77] while the dashed red line shows the treatment effect of about -0.11 log points for R$(63,70]. The treatment effect for R$(70,77] is an order of magnitude larger than any of the "treatment effects" from the placebo regressions, giving us high certainty that the number of individuals reporting incomes in R$(70,77] was impacted by the reform. Similarly, the treatment effect for R$(63,70] is larger (in absolute magnitude) than 95.7% of the placebo "treatment effects", implying a p-value of 0.043 against the null hypothesis that R$(63,70] was not impacted by the reform.

## V. Conclusion

This paper develops a sufficient statistics framework to bound the welfare impacts of reforms featuring notches. We estimate these sufficient statistics using longitudinal administrative data in the context of the 2014 reform to one of the world's largest cash transfer programs, Bolsa Família. Despite finding strong evidence of behavioral responses to the BF reform, we find that the corresponding efficiency costs are small relative to the equity benefits of increased transfers to poor households: a back-of-the-envelope calculation suggests that the welfare effect of spending R$1 on the reform is at least as high as the welfare effect of spending R$1.50 on a non-distortionary universal transfer. Because the Bolsa Família eligibility threshold is based on self-reported income, this result provides some evidence against the commonly held belief that targeting transfer programs based on self-reported incomes will generate substantial efficiency costs in high informality settings. Thus, our findings may have important implications for the future design of cash transfer programs.

Moreover, we believe that our sufficient statistics framework highlights a new manner in which reduced-form evidence on jumping and bunching can be used to inform policy. Given the ubiquity of notches, we hope that the methods developed in this paper will be useful for analyzing reforms in a variety other contexts such as Medicaid, income-dependent tax credits, or firm tax schedules. Finally, the bounding techniques developed in this paper may be helpful for bounding welfare impacts of large reforms which do not necessarily feature notches.

[37] See Imbens and Rubin (2021) or Young (2018) for discussions of randomization inference.

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